ELEC 6081 - Biomedical Signals and Systems

Dr. H. C. WU

Department of Electrical and Electronic Engineering
The University of Hong Kong

COURSE ADMINSTRATION

Lecture hour and Location

- Wed. 13:00 – 16:00 , KK202 (First Class 16/09/24)

Instructor

- Dr. H. C. Wu (andrewhcwu@eee.hku.hk)

Whatsapp

Wechat

No Textbook

Optional Reading:

- R. M. Rangayyan,
  Biomedical Signal Analysis: A Case-study Approach.
  New York: John Wiley & Sons, 2002
- J. L. Semmlow,
  Biosignal and Biomedical Image Processing: MATLAB-based Applications.
  New York: Marcel Dekker, 2004.
- L. Sornmo and P. Laguna,
  Bioelectrical Signal Processing in Cardiac and Neurological Applications.
  Burlington, MA; London: Elsevier Academic Press, 2005IT-Press, 2013.

COURSE ASSESSMENT

One Computer based assignment (10%):

- You will be provided a set of Python program codes to analyze the biomedical signals.
- There will be a tutorial on how to work out the assignment.

One written assignment (30%):

- Moodle Multiple Choice Exercise

Two-hour Examination(60%):

- Multiple Choices and Short questions (Word-based)

WORK OUT THE TUTORIAL QUESTIONS !!

JAN-26					[1]	2	3		
	4	5	6	7	8	9	10		
	11	12	13	14	15	16	17	Break	
	18	19	20	21	22	23	24	1	
	25	26	27	28	29	30	31	2	Lecture 1
FEB-26	**1**	2	3	4	5	6	7	3	
	8	9	10	11	12	13	14	4	
	15	<16>	[17]	[18]	[19]	20	21	5	University Holiday
	22	23	24	25	26	27	28		
MAR-26	**1**	2	3	4	5	6	7	6	
	8	9	10	11	12	13	14	7(Reading)	
	15	(16)	17	18	19	20	21	8	University Holiday
	22	23	24	25	26	27	28	9	
	29	30	31					10	
APR-26				1	2	[3]	[4]		
	5	[6]	[7]	8	9	10	11	11	Public Holiday
	12	13	14	15	16	17	18	12	
	19	20	21	22	23	24	25	13	Written assignment deadline
	26	27	28	29	30			14	Revision
						[1]	2		

COURSE OUTLINE

Week	Lecture	Due
1 (Jan 21)	No Class	
2 (Jan 28)	Ch-1 Introduction	
3 (Feb 4)	Ch-2 Basics of Signal Processing	
4 (Feb 11)	Ch-3 Random Process and Spectral Estimation	
5 (Feb 25)	No Class (The week after Chinese New Year)	
6 (Mar 4)	Ch-4: ECG and EMG Signal Processing	
7 (Mar 11)	Reading Week (No-class)	
8 (Mar 18)	Ch-5: Pattern classification and machine learning	
9 (Mar 25)	Tutorial for Computer Based Assignment	
10 (Apr 1)	Ch-6: Multivariate Signal Processing	
11 (Apr 8)	Ch-7: Optimal and Adaptive Filtering	
12 (Apr 15)	Ch-8: EEG and EP Signal Processing	Moodle MC Exercise Due
13 (Apr 22)	Revision	Computer Based Assignment Due

6

What is Biomedical Signals and Systems?

- Biomedical Engineering can be divided into 4 major streams

Background:

Physics

Chemistry

Biology

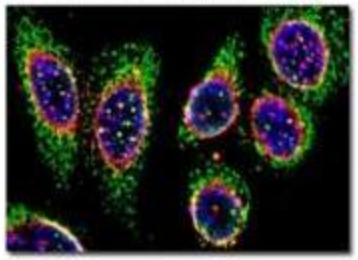
Mechanical Engineering

Biomechanics & Bio-material

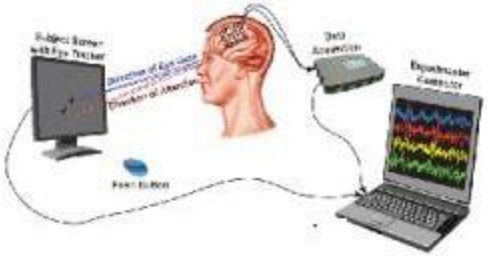
Cochlear implants

Pacemakers

Cellular Bioengineering



Biomedical System & Signals



Biomedical Inforamtics

Background:

Physics

Electronics Engineering

Machine Learning

Statistics

Data Analytics

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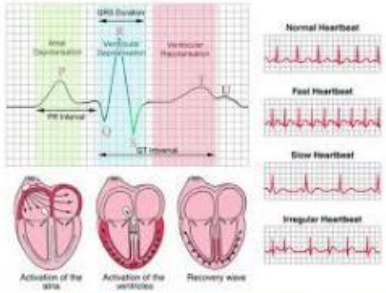
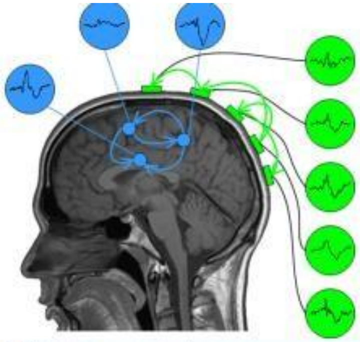
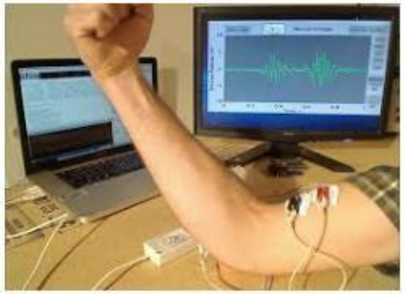
- Biomedical Signals and Systems are related to EEE.

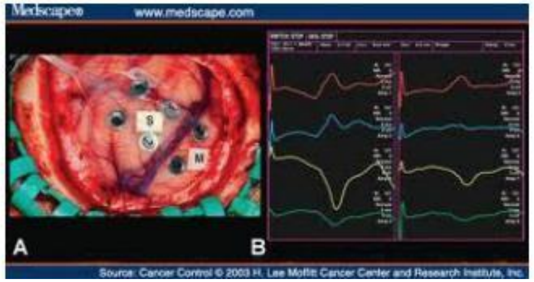
Why do we study Biomedical Signal and Systems?

- Introduction to common biomedical signals
- Signal Processing and machine learning skills
- A taste of Analyzing signals using MatLab
- Jobs !!! (Sales Engineer,Biomedical Engineer, Hospitals, etc.)







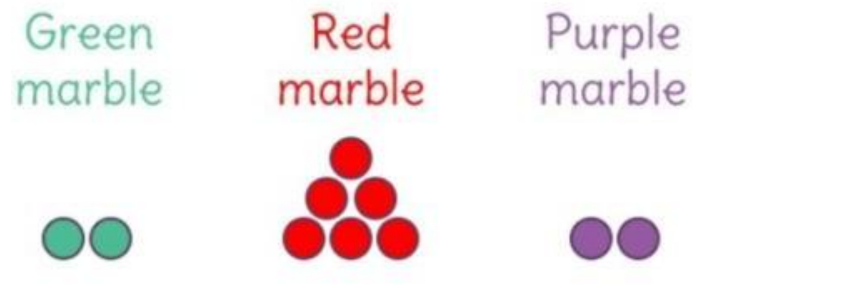


Gleneagles HONG KONG 港怡醫院
Biocare
Adventist Health 港安 香港港安醫院 荃灣
Medisource Supply (Asia) Co. Ltd
GE Healthcare
novoheart
EG GILERO
iDSMED
mindray

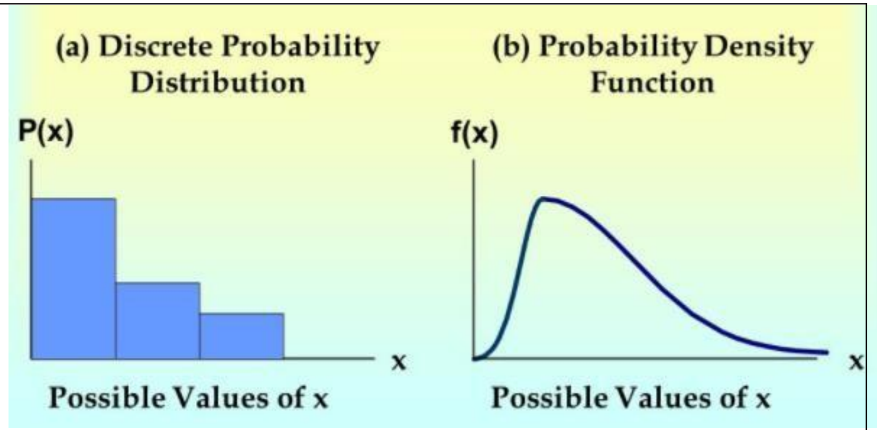
Pre-requisite

Required:

- **Basic probability and statistics**



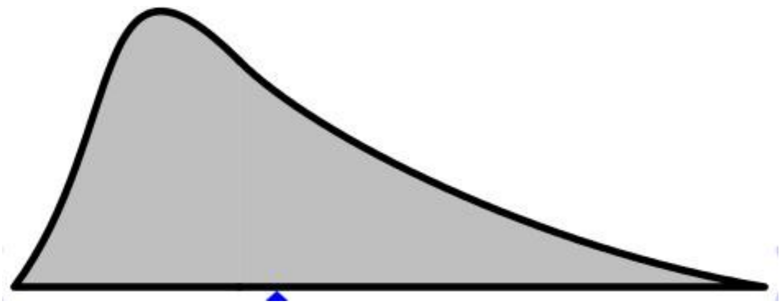
P(Green)=2/(2+6+2)=0.2



Expected value (Weighted mean)

80	20%
50	20%
100	60%

$$\bar{X}_w = \frac{\sum w_i x_i}{\sum w_i}$$



$$E[g(X)] = \int_{-\infty}^{+\infty} g(x) p_X(x) dx.$$

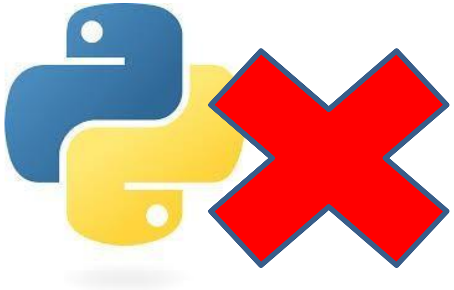
Pre-requisite

- **Basic linear algebra (Vector , matrix multiplication)**

$$\underbrace{\begin{bmatrix}1 & 2 & 3\end{bmatrix}}_{1\times 3} \cdot \underbrace{\begin{bmatrix}2 & 1 & 3\\ 3 & 3 & 2\\ 4 & 1 & 2\end{bmatrix}}_{3\times 3} = \begin{bmatrix}1\cdot 2+2\cdot 3+3\cdot 4\\ 1\cdot 1+2\cdot 3+3\cdot 1\\ 1\cdot 3+2\cdot 2+3\cdot 2\end{bmatrix} = \underbrace{\begin{bmatrix}20\\ 10\\ 13\end{bmatrix}}_{1\times 3}$$

$$\boldsymbol{a}^T\boldsymbol{x} = \sum_{i=1}^{n} a_i x_i$$

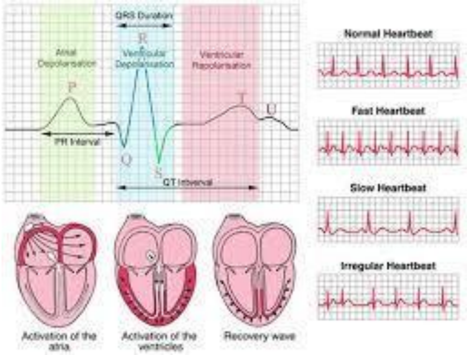
- **No background in PYTHON is assumed.**



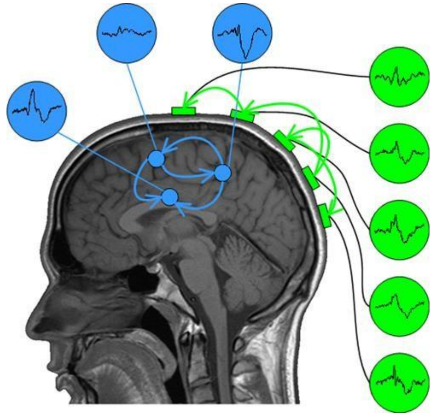
- **PYTHON Is used only in the assignment and its basic operations will be supplemented during lecture**

Course Learning Outcomes

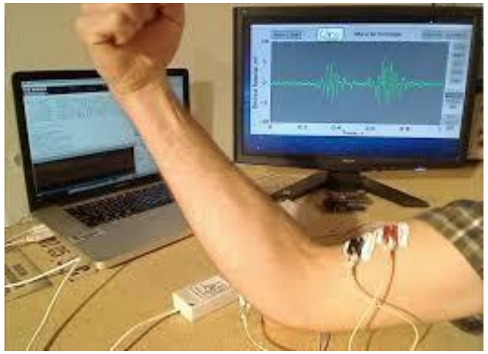
- **Master the essential knowledge of the most common and important biomedical signals and their clinical applications**



Electrocardiogram (ECG)



Electroencephalography (EEG)



Electromyography (EMG)

Course Learning Outcomes

- **Appreciate the importance of biomedical signals and systems in healthcare, physiology, and neuroscience.**

Course Learning Outcomes

- **Master application-oriented signal processing and pattern recognition techniques for the analyses of biomedical signals and systems**

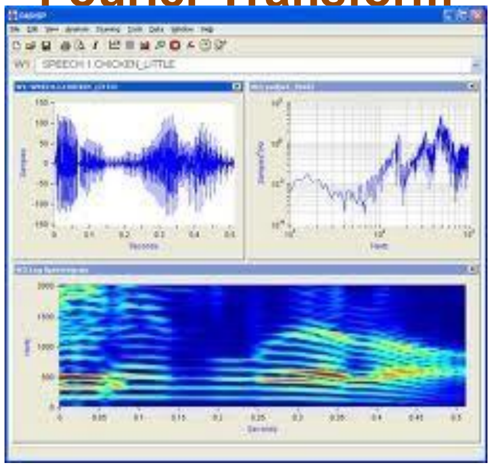
$DFT(FFT)$:

$$X(k)=\sum_{n=0}^{N-1} x(n)\cdot e^{-j\left(\frac{2\pi}{N}\right)nk} \quad (k=0,1,\ldots,N-1)$$

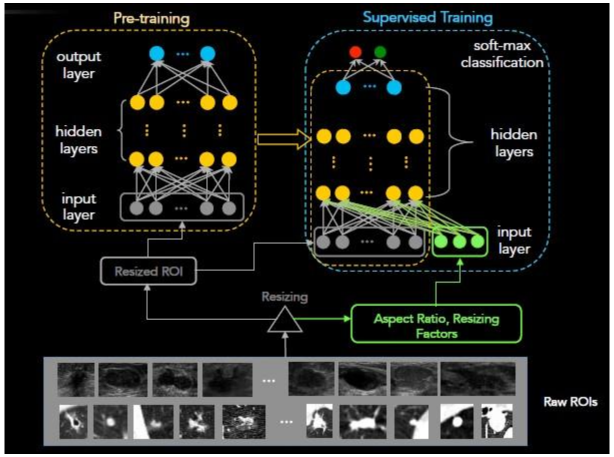
$IDFT(IFFT)$:

$$x(n)=\frac{1}{N}\sum_{k=0}^{N-1} X(k)\cdot e^{j\left(\frac{2\pi}{N}\right)nk} \quad (n=0,1,\ldots,N-1)$$

Fourier Transform



Time-Frequency Analysis



Machine Learning

Course Learning Outcomes

- **Master the ability to select appropriate signal processing techniques for real biomedical signals and to evaluate their performance in clinically relevant terms.**

Signals

ECG
EMG
EEG
ECoG
MEG
fMRI
DTI
fNIRS
Blood pressure
Respiratory
……

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Objectives

Noise reduction
Preprocessing
Signal validation
Feature extraction
Data compression
Segmentation
Pattern recognition
Trend detection
Event detection
Decision making
……

Mapping

Methods

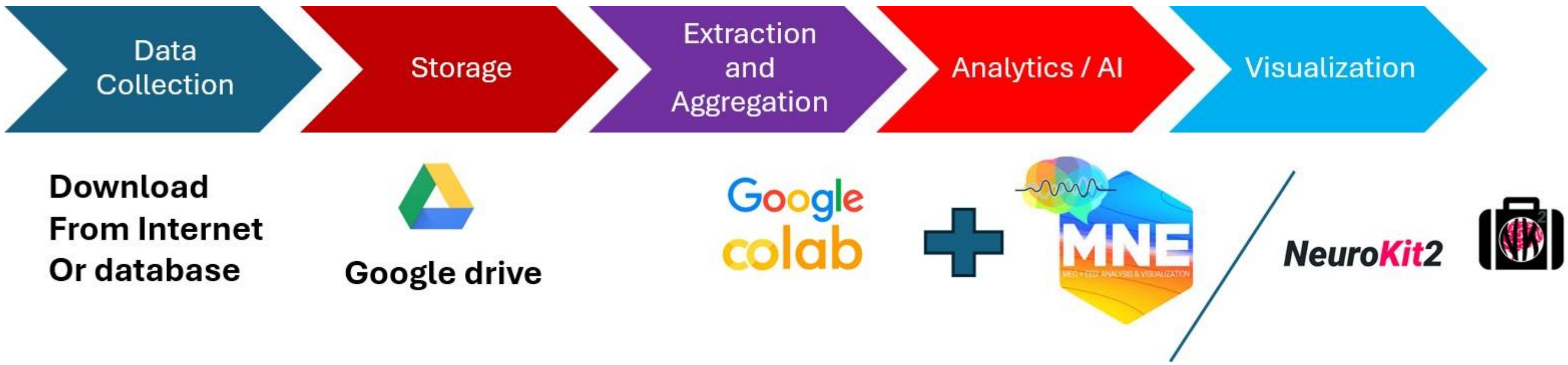
Filtering (linear, nonlinear, adaptive, optimal)
Frequency domain analysis
Time-frequency analysis
Wavelets and filter banks
Fuzzy logic
Artificial neural networks
Expert systems
State-space modelling
PCA, ICA, SVD
Clustering
LDA, SVM
Nonlinear dynamics
Complexity and fractals
……

Topics not covered

- Biomedical Instrumentation and Medical Devices
- Magnetic Resonance Imaging, Ultrasound Imaging, Optical Imaging
- Gene Data Analysis
- Animal Study

BIOMEDICAL SIGNAL PROCESSING TOOLS/SOFTWARE

- **PYTHON (Can use Google Colab)**



- **An internet browser and google drive are all you needed.**
- **Alternatively, can install python (e.g. anaconda) in your PC.**

INTRODUCTION AND OVERVIEW

CONTENTS

- OVERVIEW OF BIOMEDICAL SIGNAL AND SYSTEMS
- INTRODUCTION TO COMMON BIOMEDICAL SIGNALS
  - ELECTROCARDIOGRPAHY (ECG)
  - ELECTROMYOGRPHAY (EMG)
  - ELECTROENCEPHALOGRPAHY (EEG)
  - OTHER BIOMEDICAL SIGNALS
- DIFFICULTIES IN BIOMEDICAL SIGNAL PROCESSING

1. OVERVIEW OF BIOMEDICAL SIGNAL AND SYSTEMS

1. Definition

- **Biomedical Signal**
  - 1D waveforms
  - E.g. ECG (Heart), EEG(brain)

- **Biomedical Images**
- Images/Videos
- E.g. Ultrasound, MRI

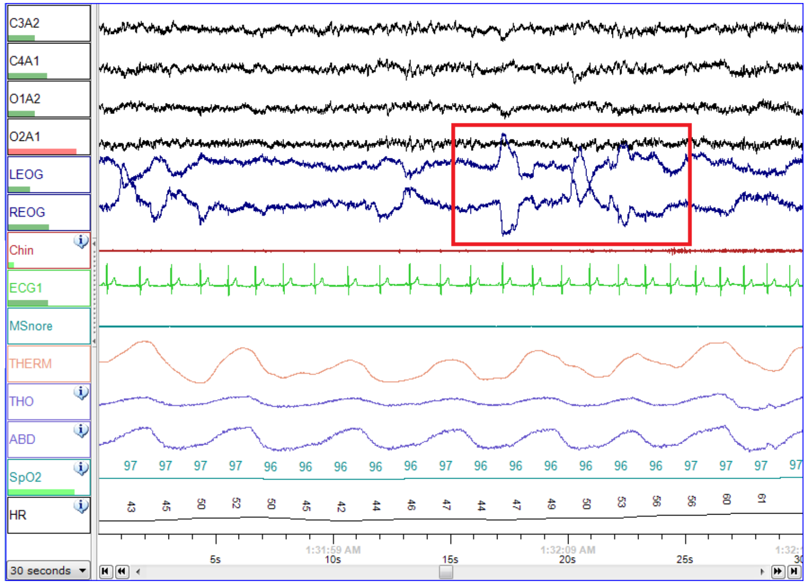


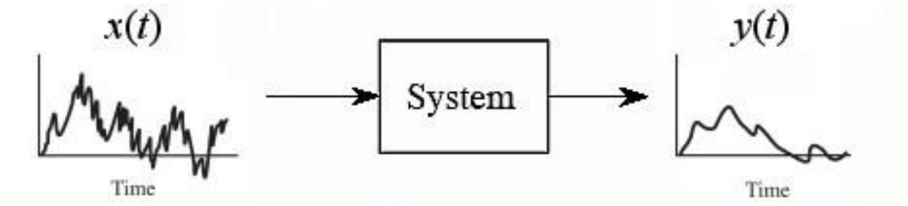
Fig. 1(a) PSG signal

Fig. 1(b) Ultrasound signal

Biomedical System

– A set of interacting components integrating into a structure with certain behavior

1. Physiology system: systems regulating different human organs to stay alive
2. Measurement system: Acquire signals
3. Information management system



Different levels of physiological systems

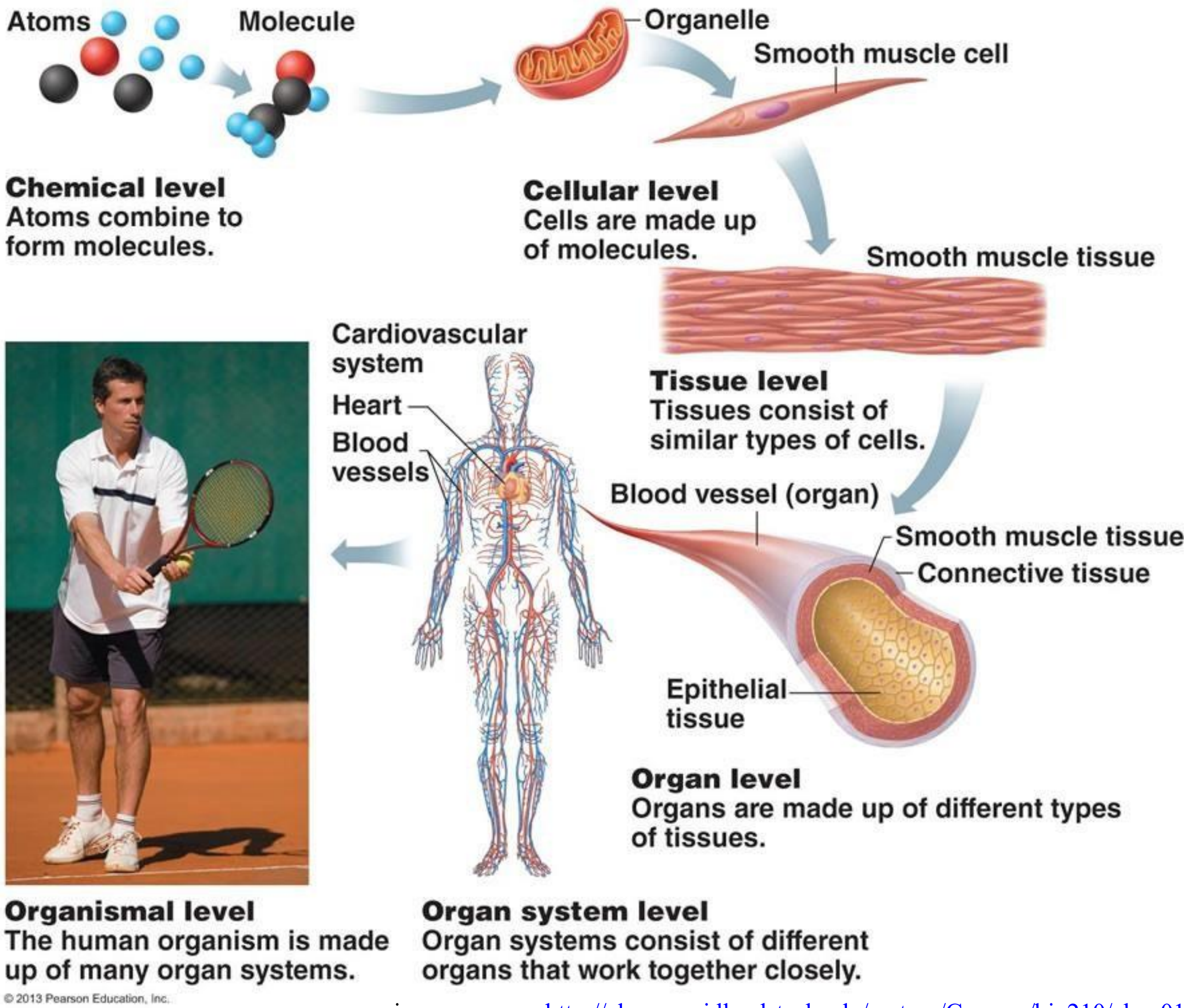
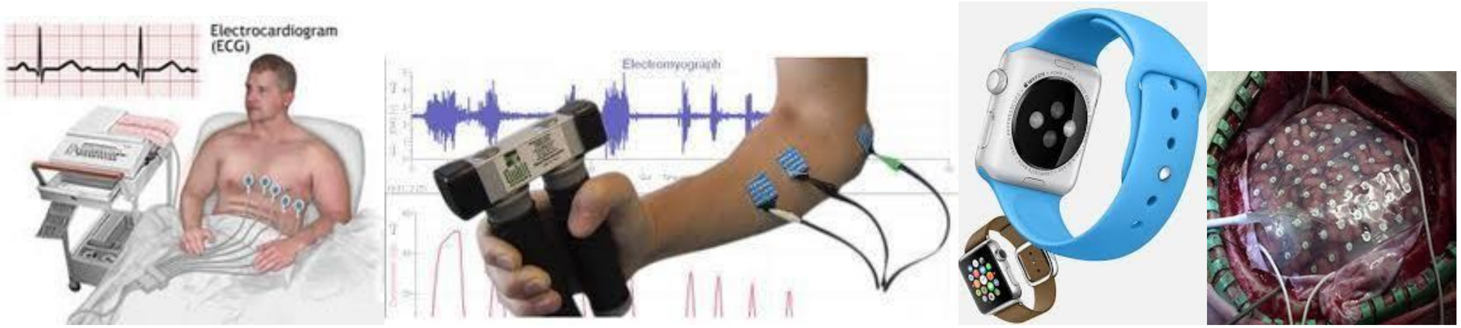


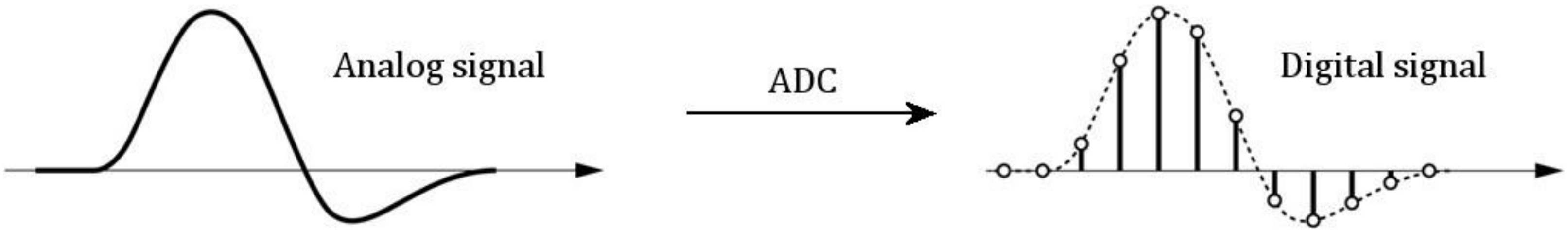
image source: http://classes.midlandstech.edu/carterp/Courses/bio210/chap01/chap01.html

1.2 General Properties of Biomedical Signals

- **Signal Acquisition:** by a sensor or a transducer and is converted to a proportional voltage or current for processing and storage.

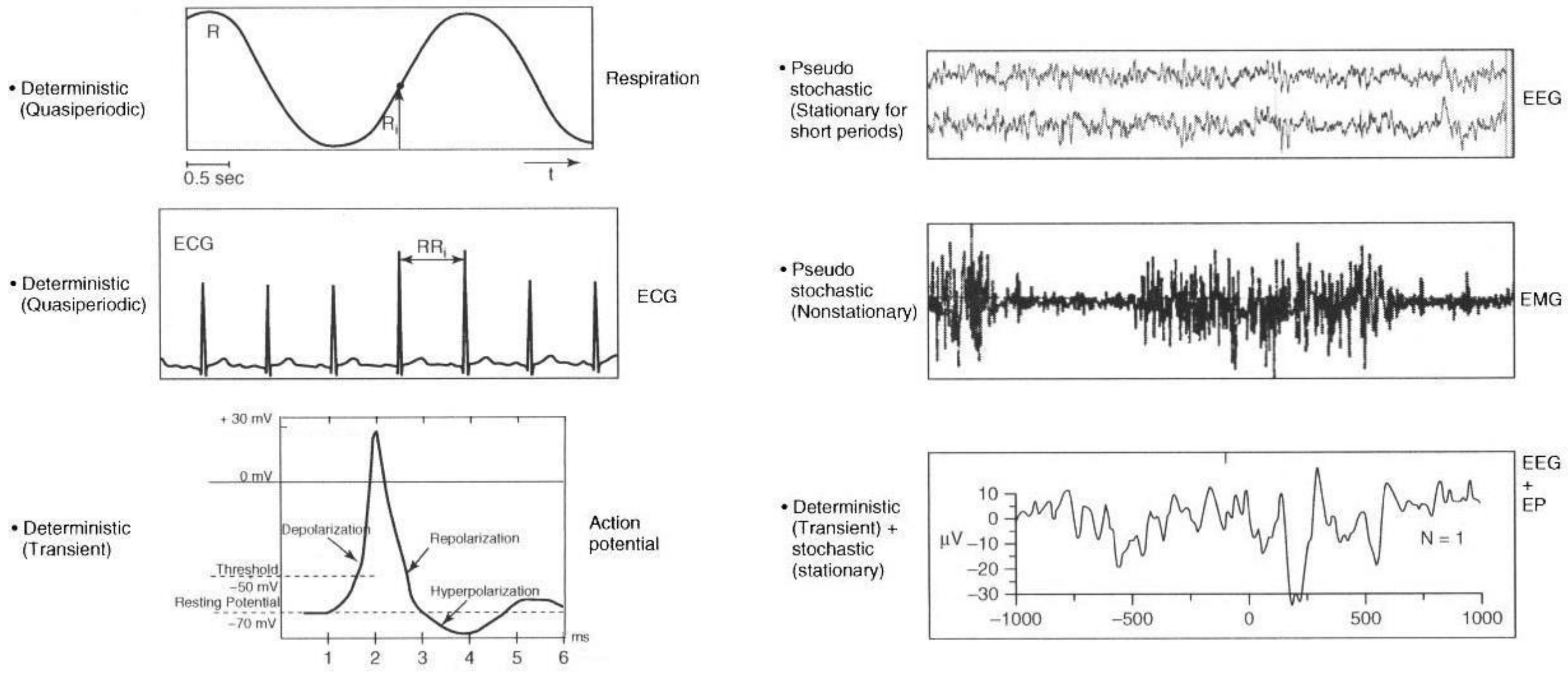


- **Characteristics:** Most are analog (I.e. continuous in time and magnitude) and is converted to digital form by analog-to-digital conversion (ADC) for storage, transmission and analysis



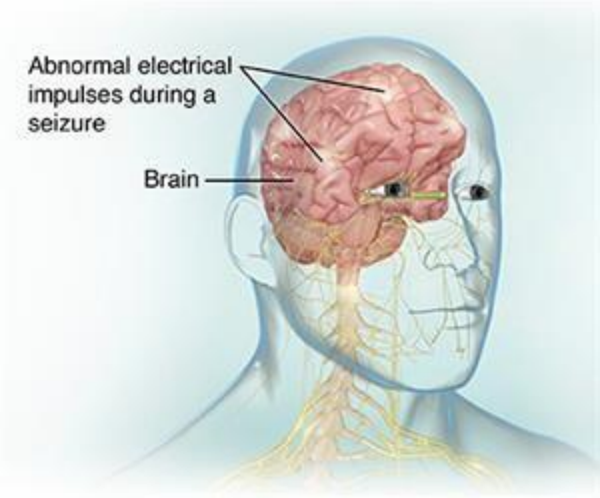
- **Deterministic Signal :** can be exactly described mathematically.
- **Stochastic Signal:** exhibits randomness, having a random probability distribution or pattern that may be analysed statistically but may not be predicted precisely.

- **Stationary:** Statistical properties do not change over time
- **Non-stationary:** Statistical properties change over time



1.3 Goal of obtaining Biomedical Signals

- **Advancement of knowledge:** To measure phenomena for understanding the underlying mechanisms
- **Monitoring:** To obtain continuous or periodic information about the system with the aim to react when necessary
- **Diagnosis:** To detect malfunction, pathology, or abnormality
- **Intervention, treatment, and rehabilitation:** To modify the behavior of the system and ensure the results
- **Evaluation:** To assess performance and effect of treatment and rehabilitation



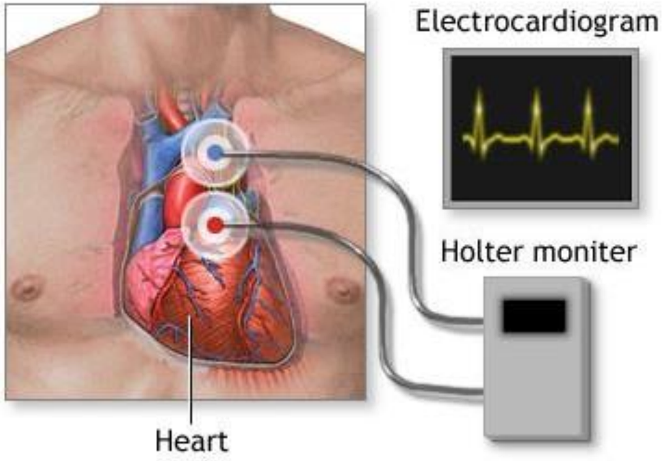


Image source :https://medlineplus.gov/ency/article/003877.htm

1.4 Types of Biomedical Signals

1. **Electrical signals:** electrical activities of the brain, the heart, muscles; recorded using electrodes
2. **Magnetic signals:** magnetic activities of the brain, the heart, lungs; recorded using magnetometers
3. **Acoustic signals:** sounds from blood flow, lungs, digestive tract, joints, muscles; recorded using microphones
4. **Mechanical signals:** blood pressure, limb acceleration, flow tension; recorded using diversity of mechanical sensors
5. **Impedance signals:** tissue composition, blood distribution; recorded using electrodes
6. **Optical signals:** blood oxygenation, heart output; recorded using fiberoptic technology
7. **Chemical signals:** blood/tissue samples; recorded using chemical laboratory tools
8. **Others:** nurse observations, annotations, alarms,…

Some common Biomedical Signals

- Electrocardiography
- Electromyography
- Electroencephalography

1. Respiratory signal: To measure the timing and breath of respiration

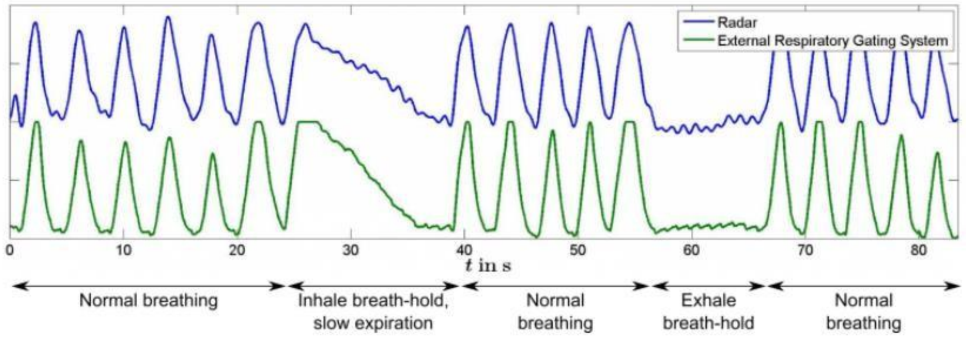
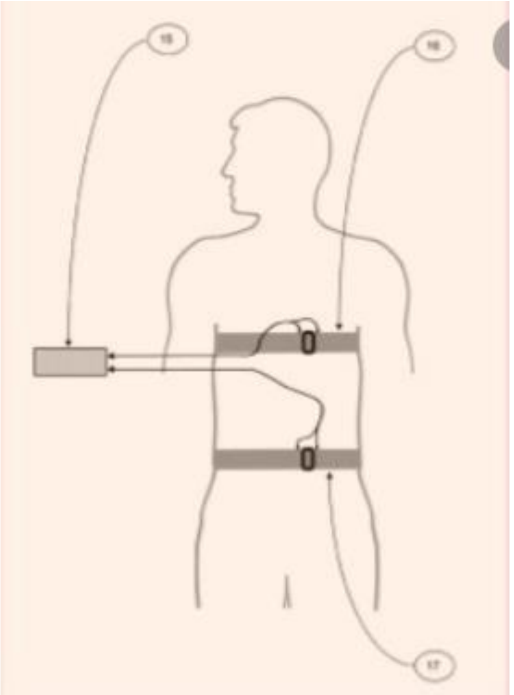


Image source: http://www.dkfz.de/en/medphysrad/workinggroups/ct/radarbased.html

Typical respiratory signals:

1. Nasal pressure,
2. thermal sensor,
3. Respiratory inductive plethysmography: measures movement of the chest and abdominal wall
4. oxygen content



Respiratory inductive plethysmography

2. Blood Pressure: To measure pressure exerted upon walls of blood vessel

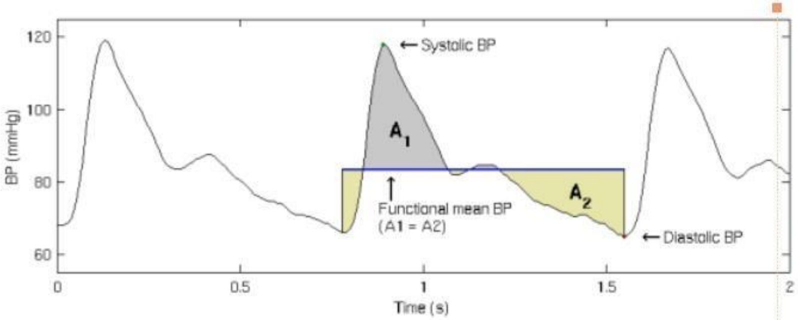
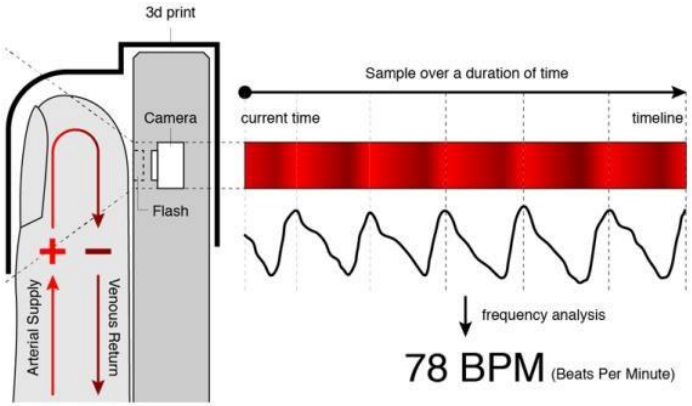


Image source: http://bsamig.uef.fi/research/bloodpress.shtml

3. Photoplethysmography (PPG): detect blood volume changes using LED at skin surface (non-invasive)



4. Otoacoustic emission (OAE): Measuring sound generated from inner ear (used for checking ear problem in newborn babies)

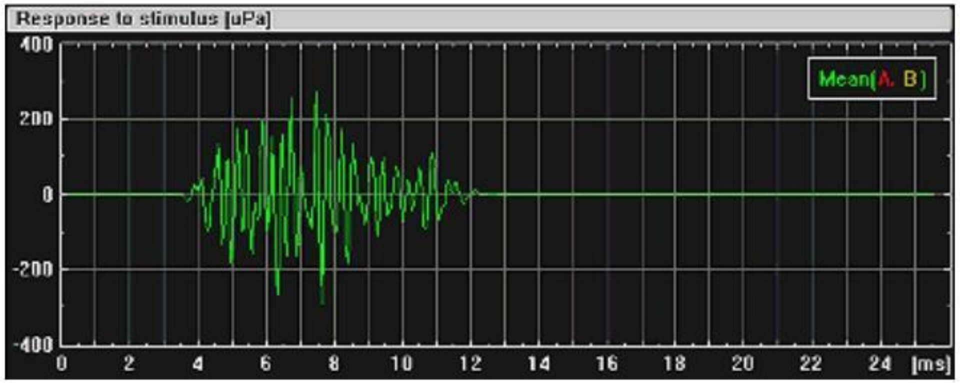
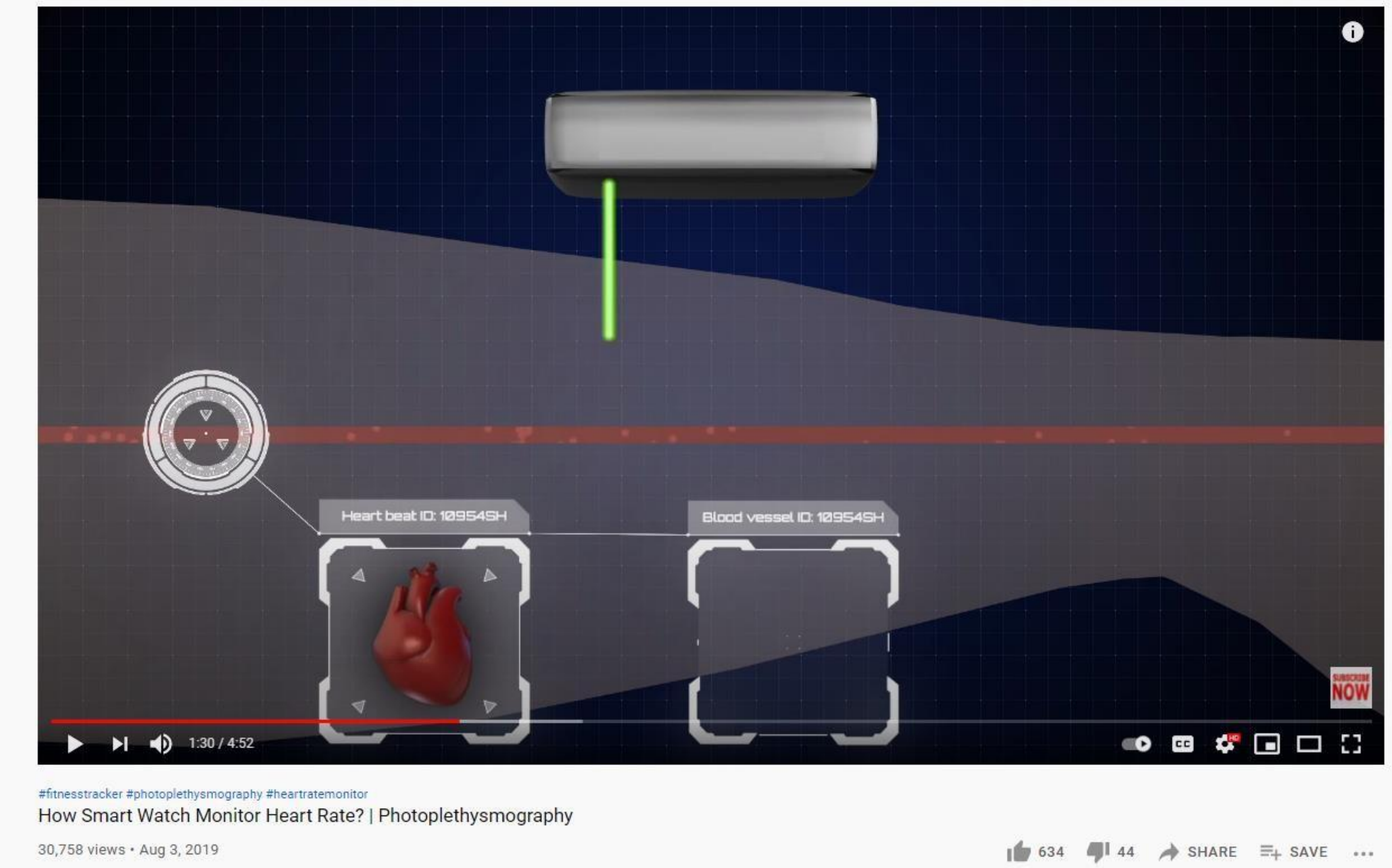
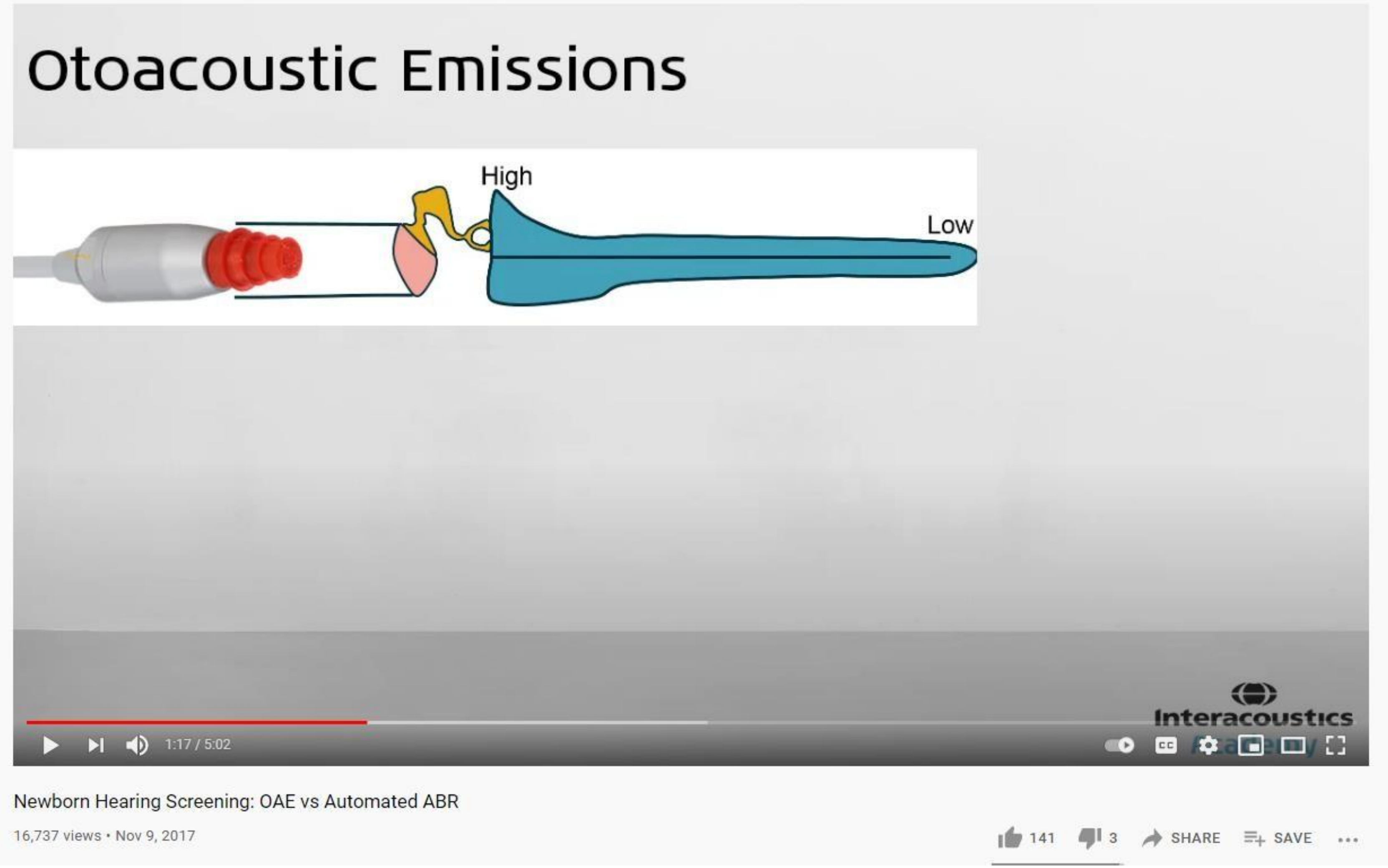


Image source: http://www.otoemissions.org/



https://www.youtube.com/watch?v=QTniLL0sQtA&ab_channel=Dr.YoDr.Yo

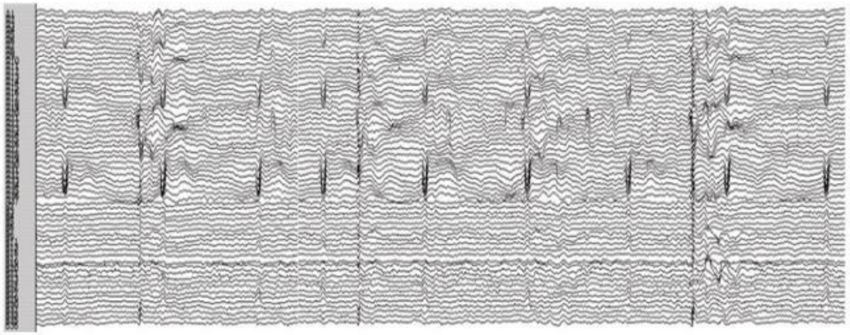


https://www.youtube.com/watch?v=5gEEE2Vf-cg&ab_channel=Interacoustics

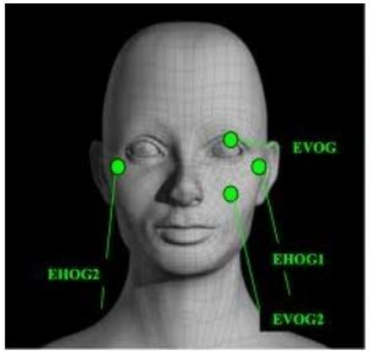
5. Magnetoencephalography (MEG):

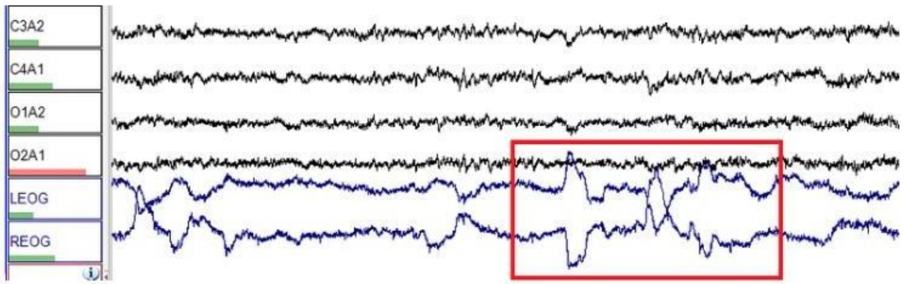
a) measuring the changes in the magnetic field caused by the activities of the brain neurons

b) with higher spatial resolution (up to about 600 sensors) and wider frequency range (from DC to >600 Hz)



6. Electrooculography : measuring the resting potential of the retina in eye





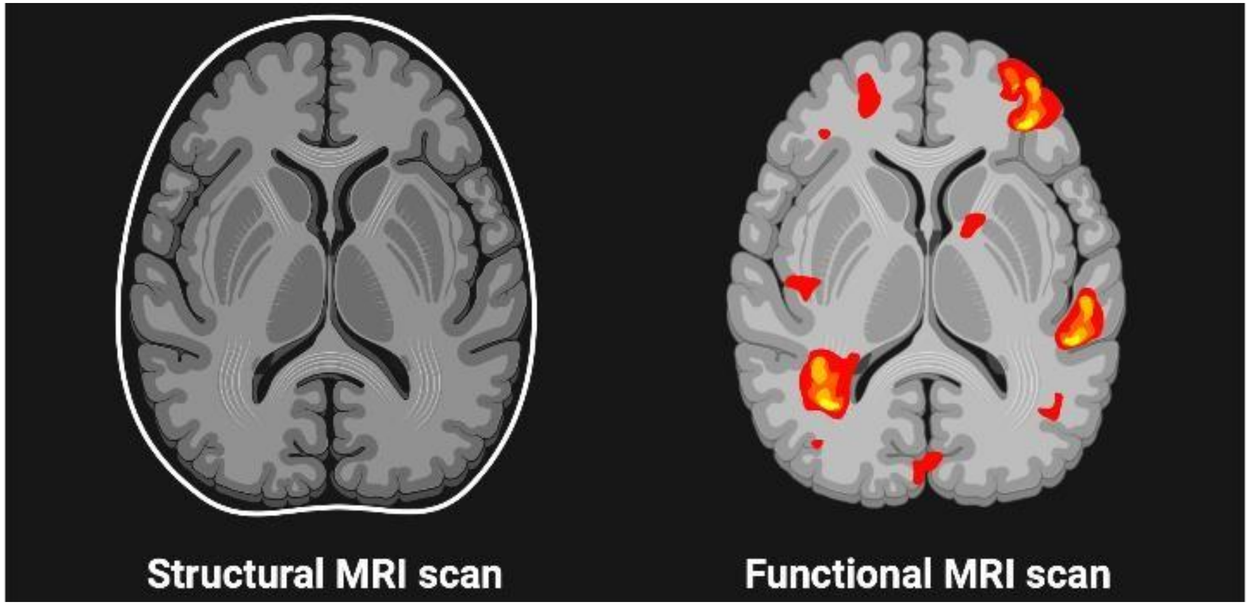
7.Magnetic resonance images: Explore the magnetic properties of water molecules to produce high-quality images

a) **Structural MRI**

- **measure of magnetic resonances of protons. Reveals brain structure/ anatomy**

b) **Functional MRI**

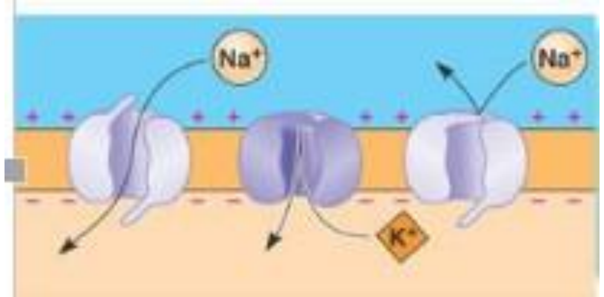
- **measure of changes in blood flow, reveals brain function.**
- **When neurons are active, they increase their consumption of energy and thus the blood flow is increased**



1.5 Electrophysiological Signals

- **Electrophysiological signals are of immense importance because they are routinely recorded in modern clinical practice.**

- **Electrocardiography (ECG): heart signals**
- **Electromyography (EMG): muscle signals**
- **Electroencephalography (EEG): brain signals**



- Others: EOG, EGG, ERG, ENG, etc.

What is Action Potential?

- origin of all electrophysiological signals
- a transient event in which the electrical membrane potential of a cell rapidly rises and falls, following a consistent trajectory
- Caused by flow of ions across cell membrane

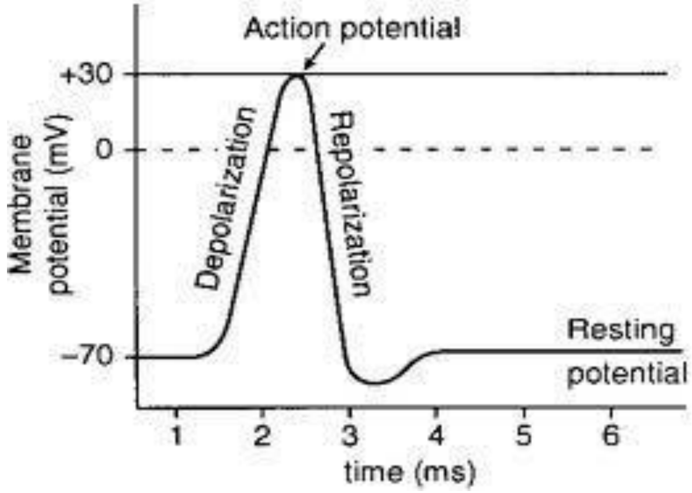
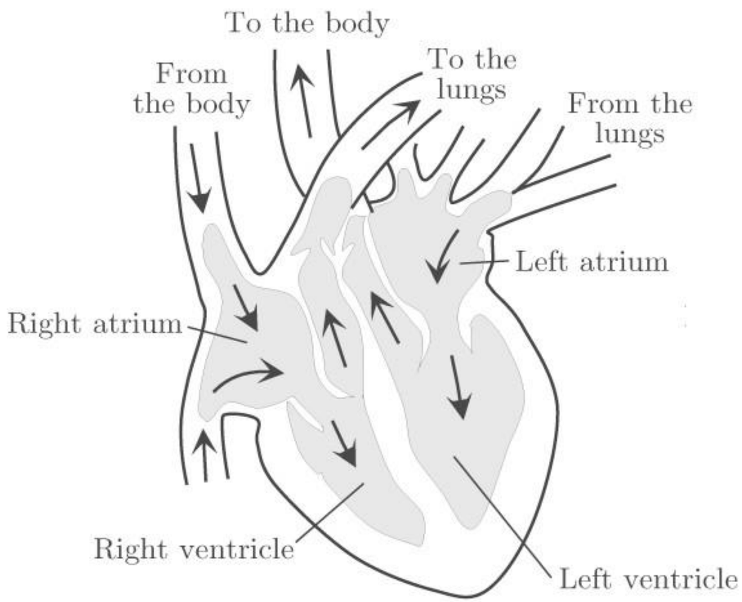
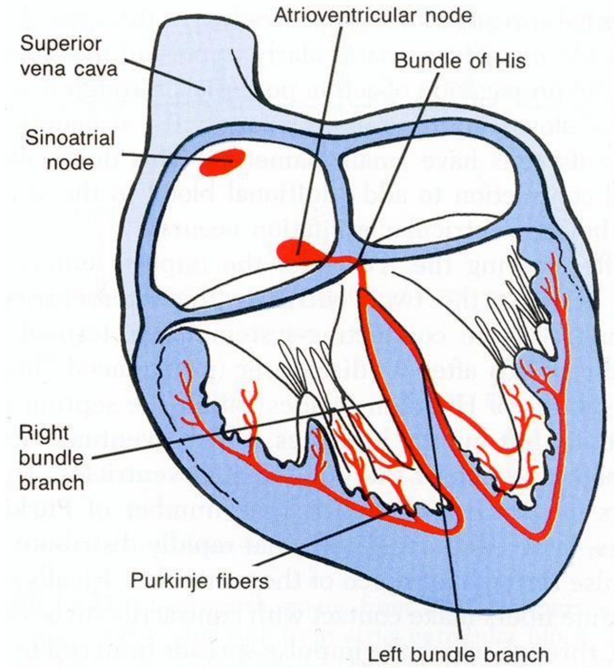


image source: http://kvhs.nbed.nb.ca/gallant/biology/action_

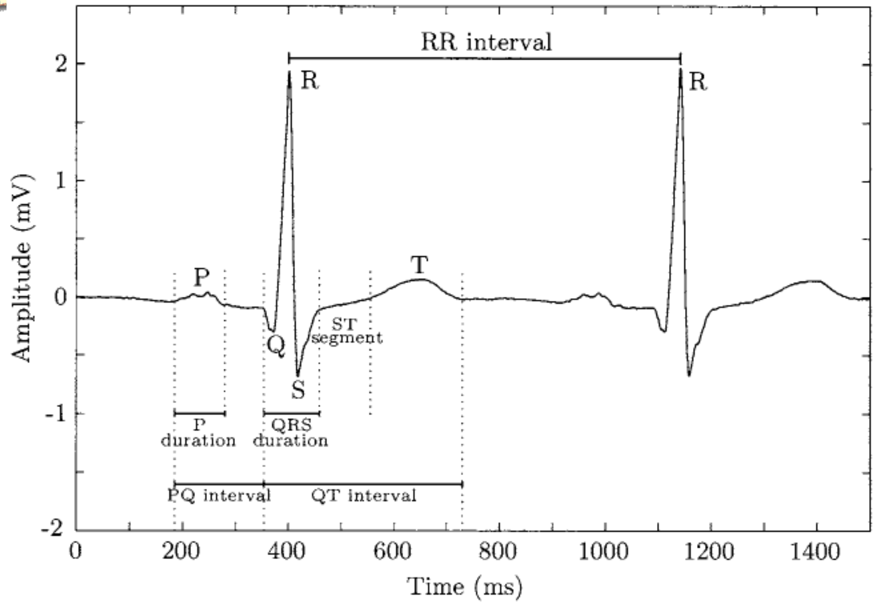
Further reading:
Section 1.2 in Sörnmo and Laguna, Bioelectrical Signal Processing in Cardiac and Neurological Applications,

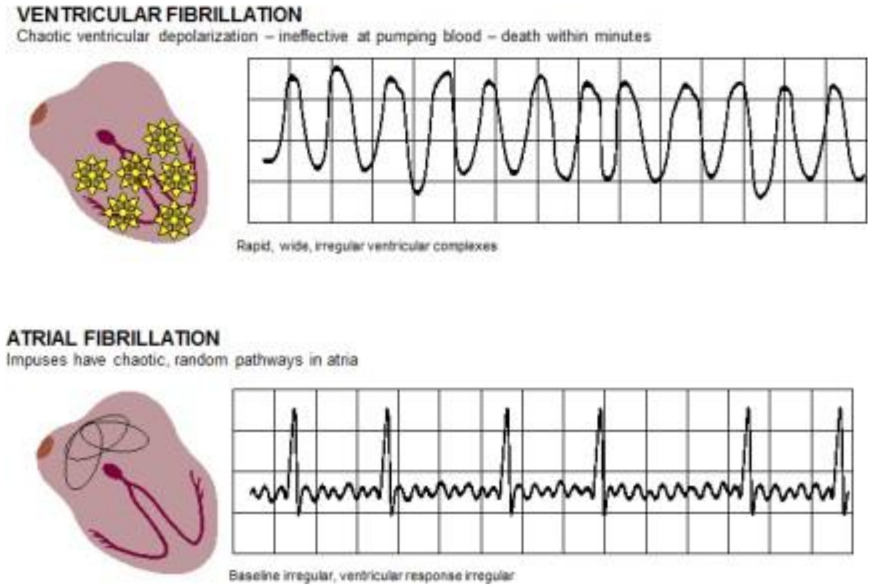
- ECG: Electrical manifestation of the contractile activity of the heart
- The heart:
  - has 4 chambers: 2 atria and 2 ventricles;
  - Uses muscle cells to contract;
  - Cardiac pace (Heart Rhythm) is created by sinoatrial (SA) node ;





Further reading:
Section 6 in Sörnmo and Laguna, Bioelectrical Signal Processing in Cardiac and Neurological Applications,





Normal Cardiac Cycle

Abnormal Cardiac Cycle

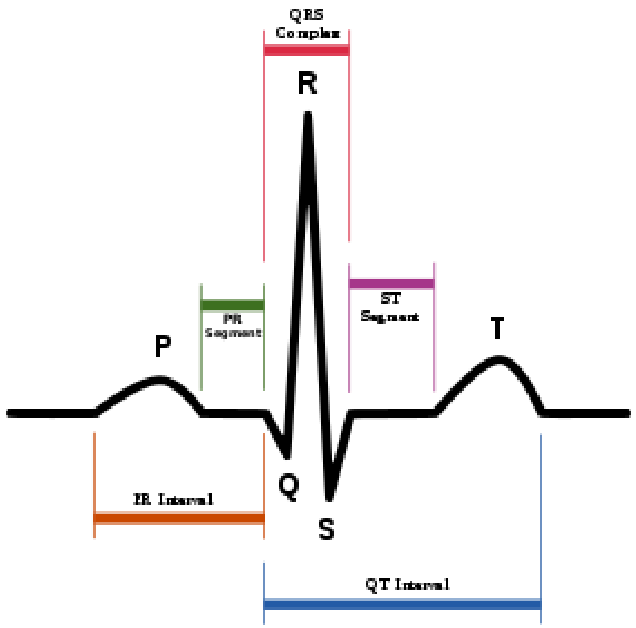
Image source: Sörnmo and Laguna, Bioelectrical Signal Processing in Cardiac and Neurological Applications, 2005

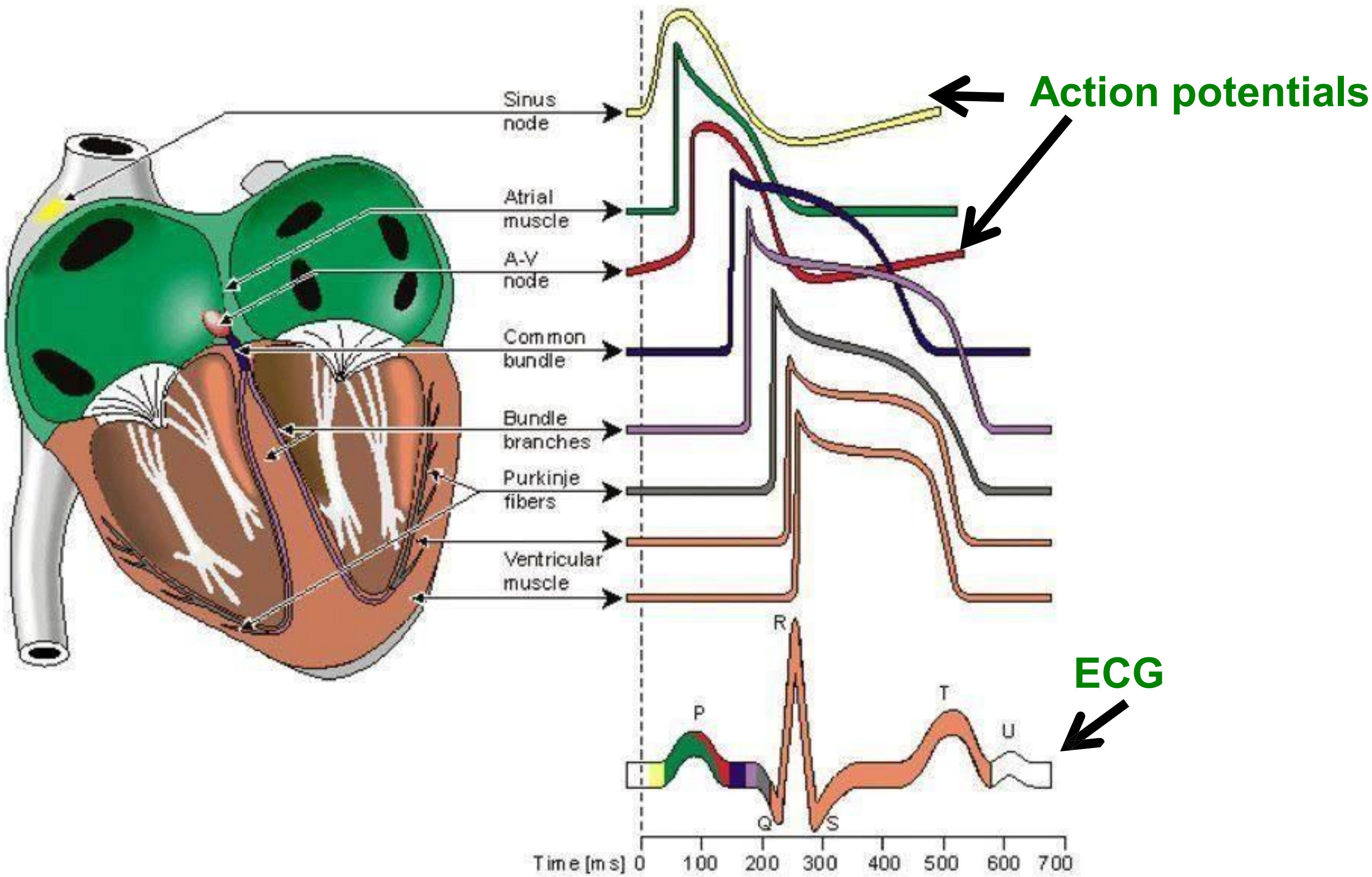
Applications:

- The ECG is often used for identifying heart problems/diseases.

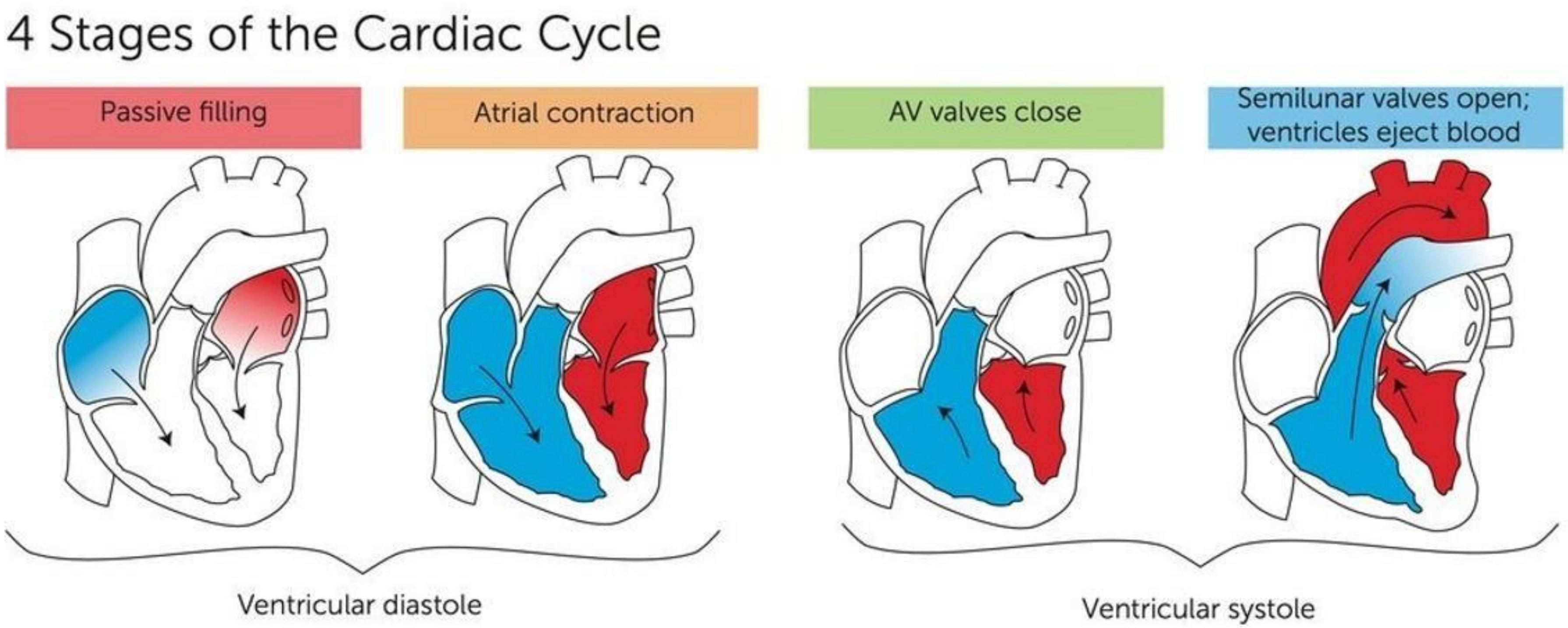
■ The ECG of a cardiac cycle can be decomposed into 7 major components.

1. **P wave:** the sequential depolarization of the right and left atria.
2. **QRS complex:** depolarization of the right and left ventricles
3. **T wave:** ventricular repolarization after the QRS complex
4. **ST segment:** the interval during which the ventricles remain in an depolarized state
5. **RR interval:** the length of a ventricular cardiac cycle
6. **PQ interval:** the time from the onset of atrial depolarization to the onset of ventricular depolarization
7. **QT interval:** the time from the onset of ventricular depolarization to the completion of ventricular repolarization



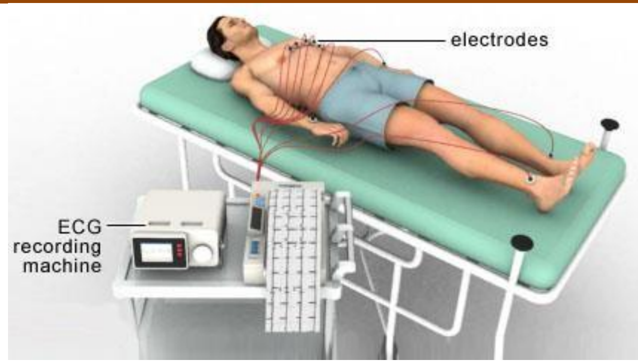


- Different ECG components are highly correlated to the Action potentials of different phases. https://www.youtube.com/watch?v=RYZ4daFwMa8

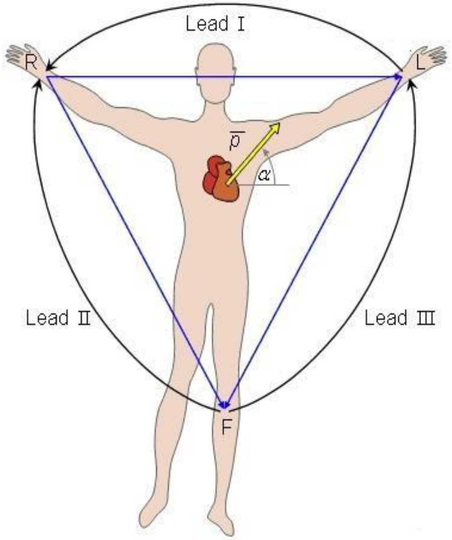


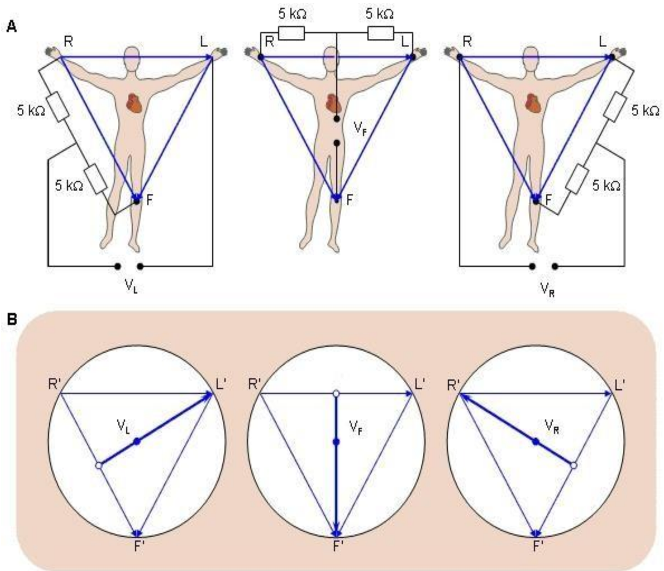
NOT EXAMINED

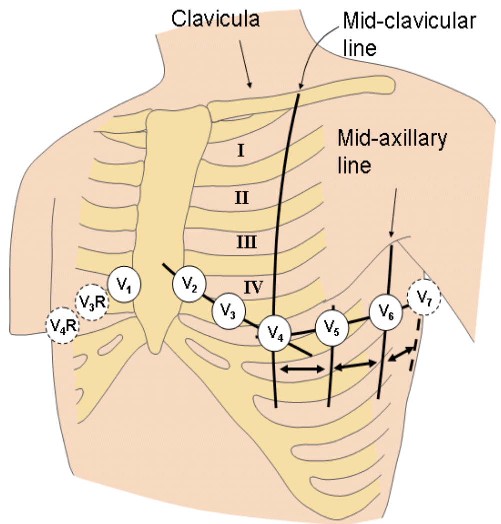
- **Lead:** the difference between two electrodes
- **Standard 12-lead system:** the most widely used, made up of the following leads



bipolar limb leads: I, II, III	augmented unipolar limb leads: aVR, aVL, aVF







12-LEAD ECG ELECTRODE PLACEMENT

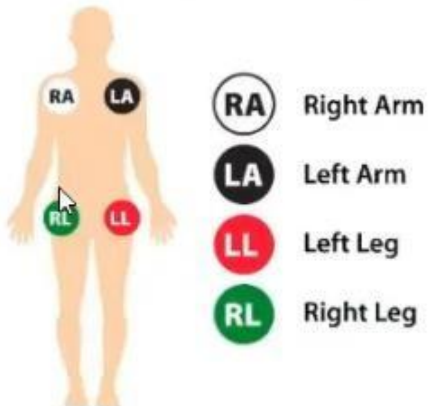
To measure the heart's electrical activity accurately, proper electrode placement is crucial.

In a 12-lead ECG, there are 12 leads calculated using 10 electrodes.

Chest (Precordial) Electrodes and Placement

- V1 - Fourth intercostal space on the right sternum
- V2 - Fourth intercostal space at the left sternum
- V3 - Midway between placement of V2 and V4
- V4 - Fifth intercostal space at the midclavicular line
- V5 - Anterior axillary line on the same horizontal level as V4
- V6 - Mid-axillary line on the same horizontal level as V4 and V5

Limb (Extremity) Electrodes and Placement



- RA (Right Arm) - Anywhere between the right shoulder and right elbow
- RL (Right Leg) - Anywhere below the right torso and above the right ankle
- LA(Left Arm) - Anywhere between the left shoulder and the left elbow
- LL (Left Leg) - Anywhere below the left torso and above the left ankle

3. ELECTROMYOGRAPHY (EMG)

- **EMG: electrical activity produced by skeletal muscles**
- **Motor units (MUs): a motor nerve cell controlling muscle fibers it innervates.**
- **action potentials (MUAP): signals controlling muscle contraction.**
- **EMG is measured as the summation of MUAPs generated by the MUs.**

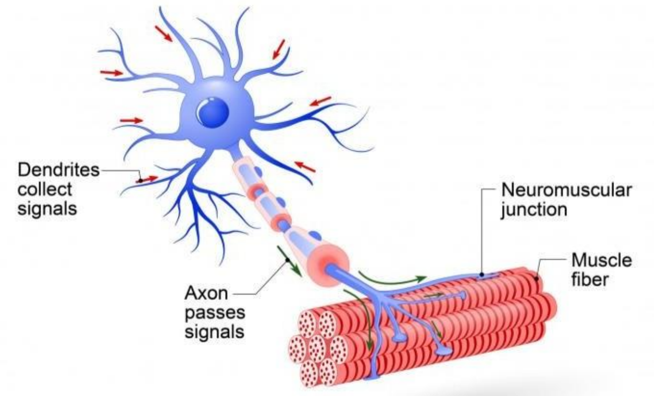
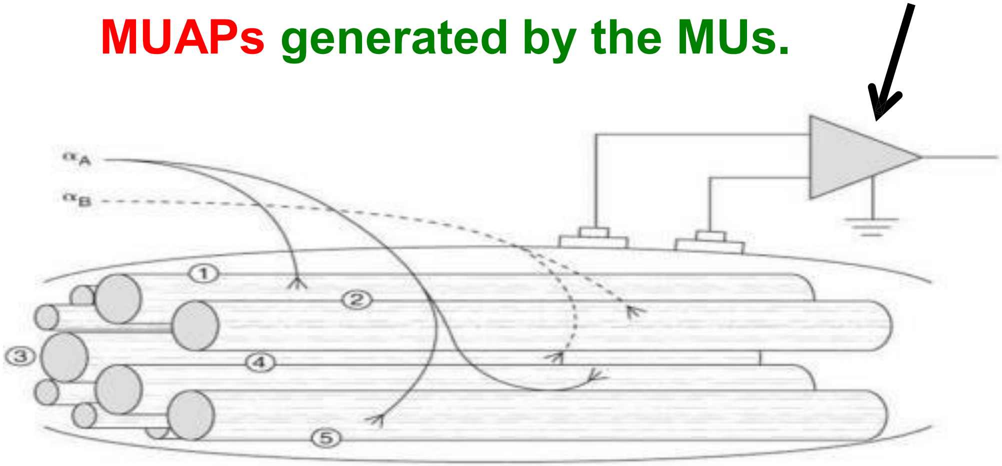
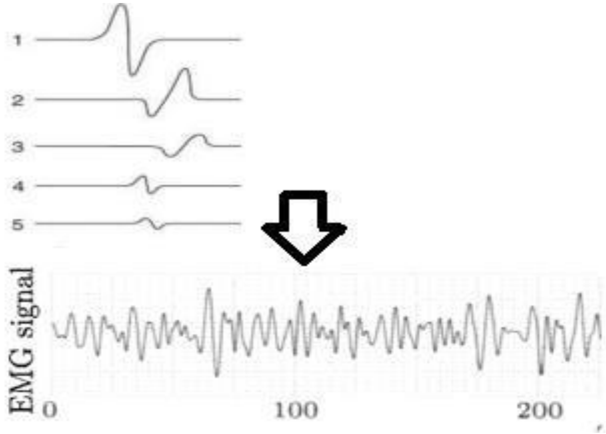


Image source: https://smanewstoday.com/wp-content/uploads/2017/09/shutterstock_423991636.jpg



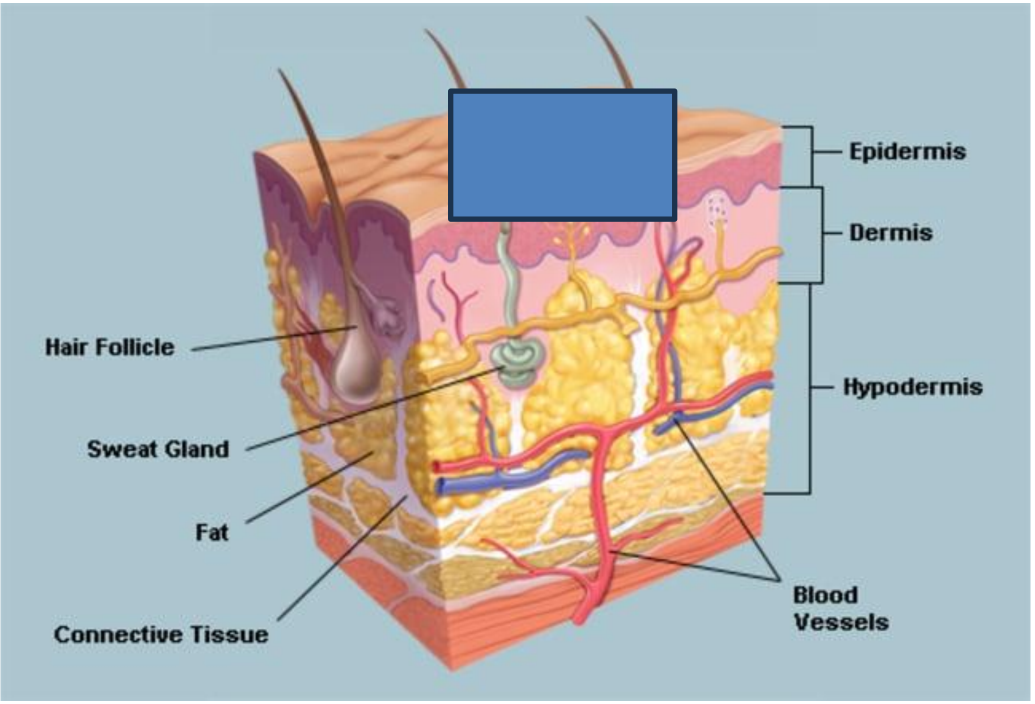


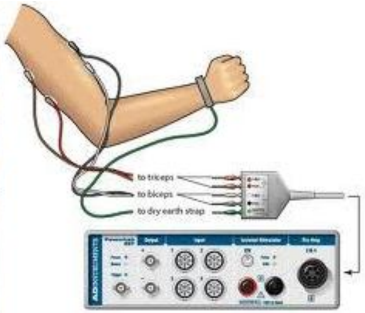
Kamen & Gabriel, 2010a, p. 11

Further reading:
Section 5.1 in Sörnmo and Laguna, Bioelectrical Signal Processing in Cardiac and Neurological Applications,2005

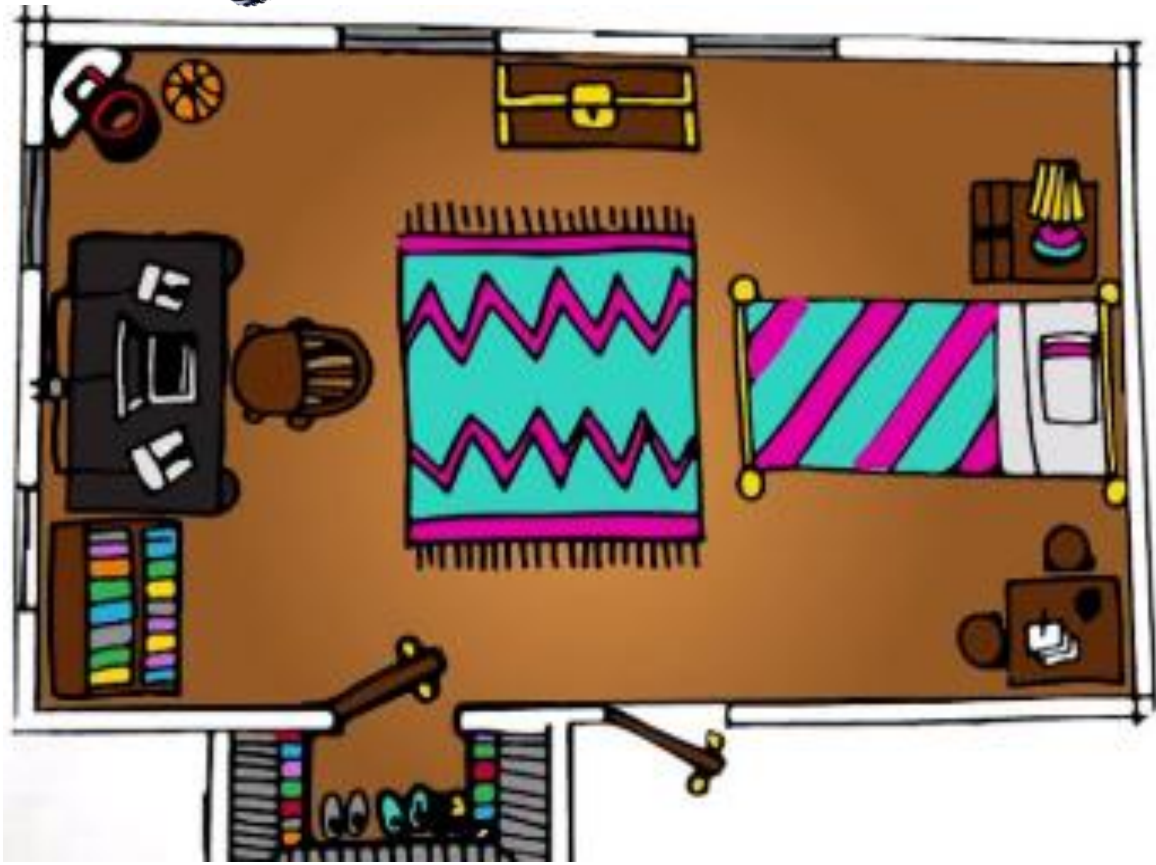
Surface EMG

Skin



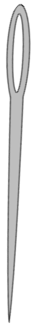


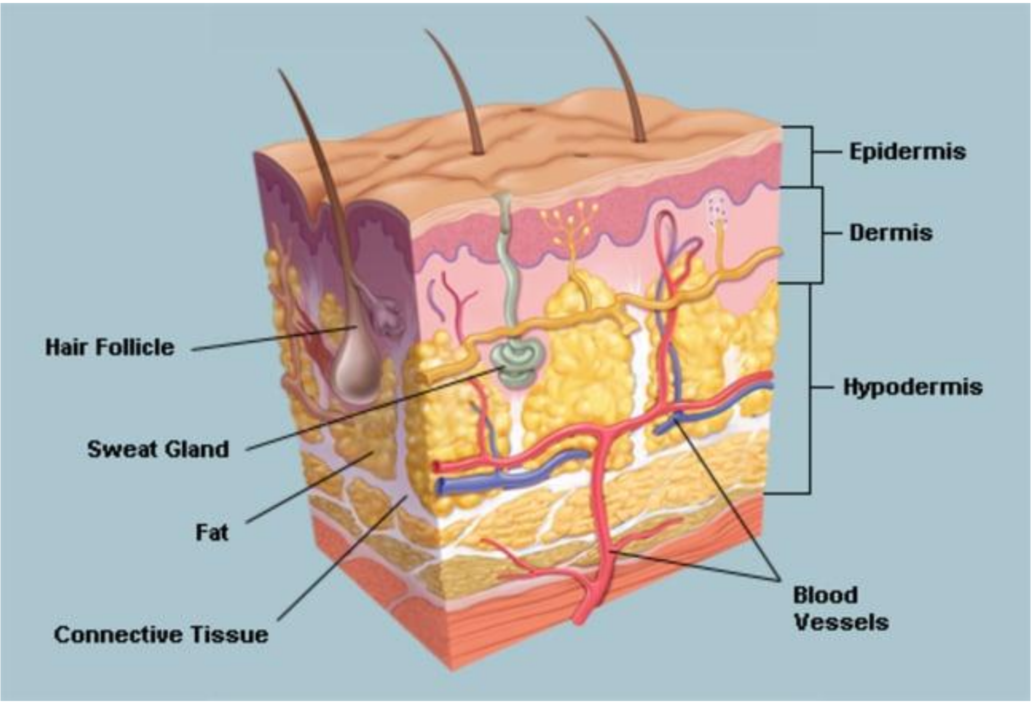
Room

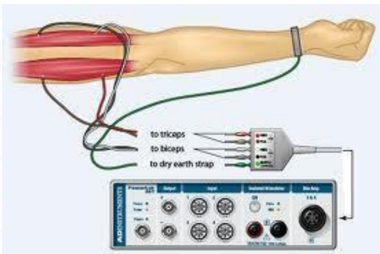


Needle EMG

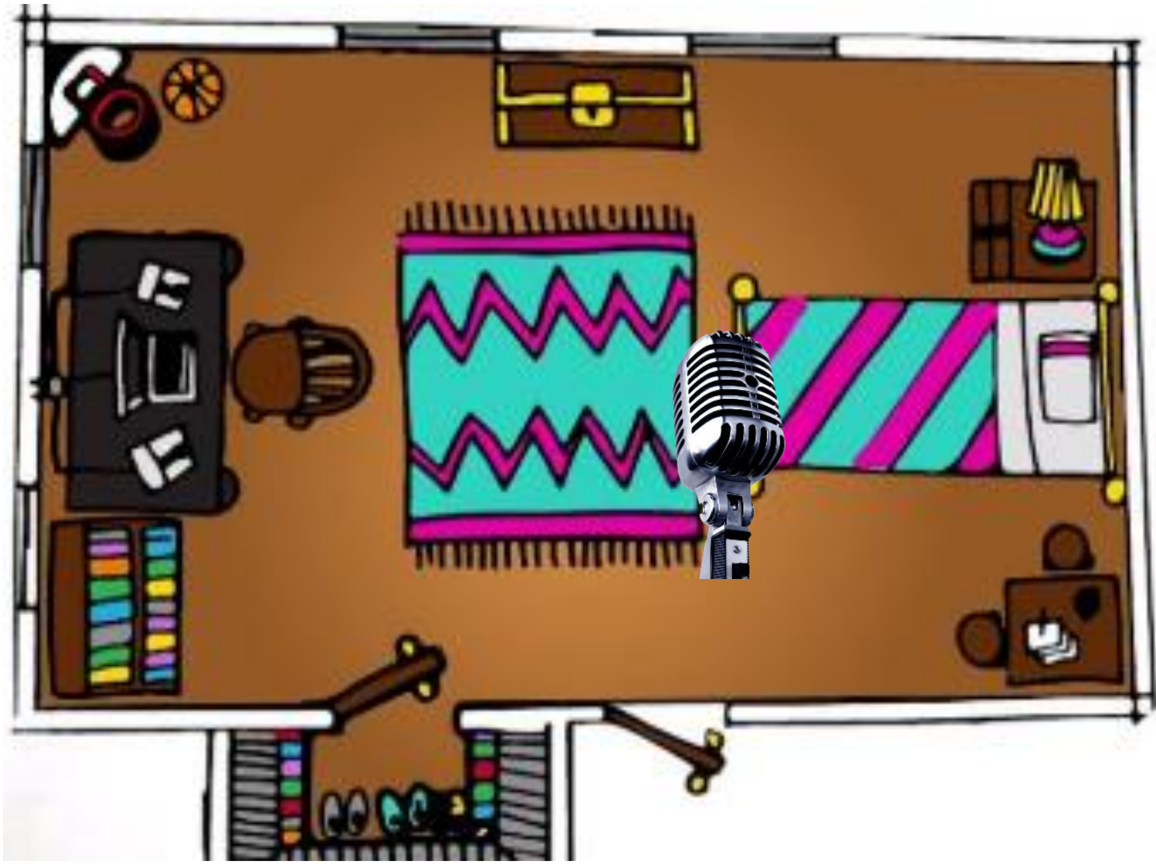
Skin





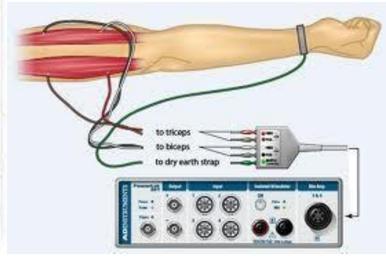


Room



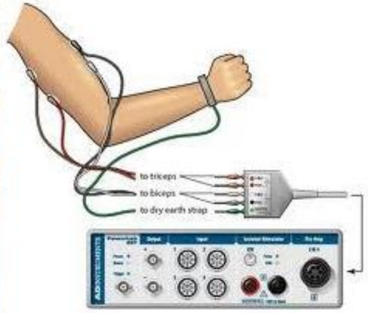
3.1 Acquisition of EMG

- **Needle EMG:** measured by inserting a needle electrode through the skin directly into the muscle



Advantages	Disadvantages
• Minimum artifacts • Allow deep muscle measurable • High spatial resolution • High signal-to-noise ratio	• Subjects may feel painful when inserting the needles • Invasive • This test must be carried out by physician

- **Surface EMG (sEMG):** measured by placing electrode on the skin over the muscle



Advantages	Disadvantages
● Non-invasive	● Cannot detect small and deep muscles
● Dynamic	● Low spatial resolution
● Electrodes are readily obtainable	• Low signal-to-noise ratio
● Lessen discomfort	

Applications:

Diagnosis, Kinesiology, Ergonomics, Prosthesis control

Kinesiology is the study of body movement.

Ergonomics: The study of muscle fatigue is central in ergonomics due to the natural wish to avoid fatigue in work situations

Prosthesis control: Control of fake arm, legs..etc..

4. ELECTROENCEPHALOGRAPHY (EEG)

- **EEG: the recording of electrical signal of a large number of neurons having similar spatial orientation**
- **Mechanism: Dendrites detect stimuli / forward nerve impulses (I.e. action potentials)**

Fig. 1. Common brain regions and functions

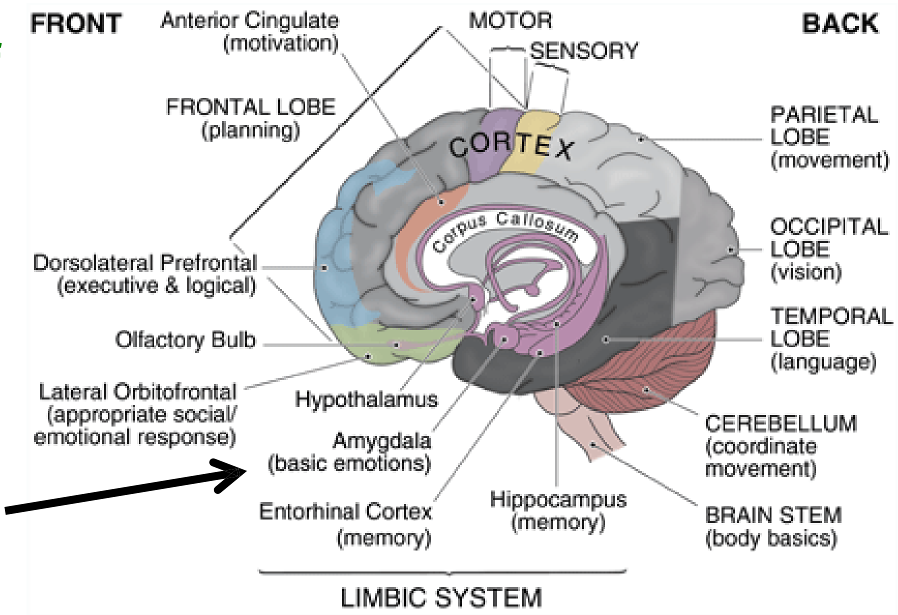
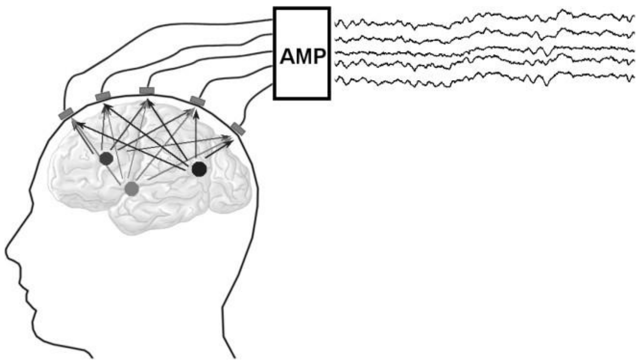


Image source: http://www.brainwaves.com



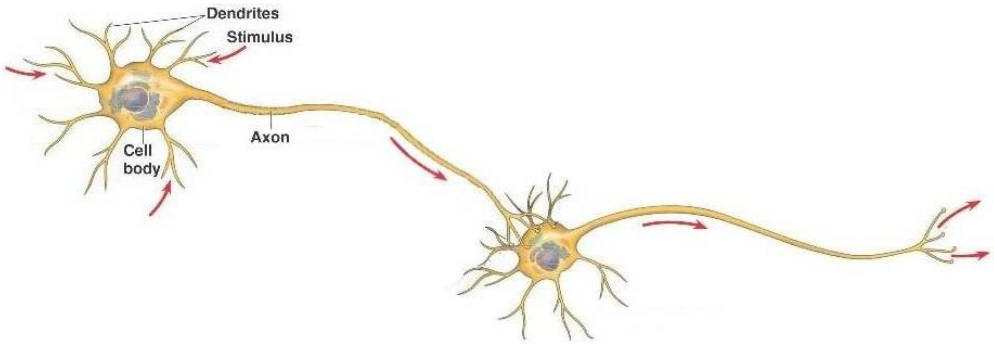
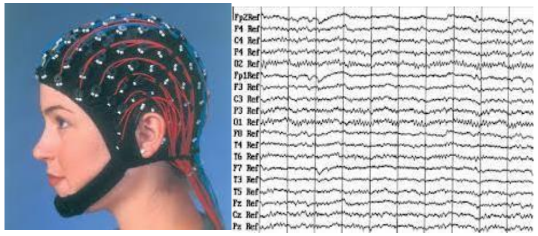
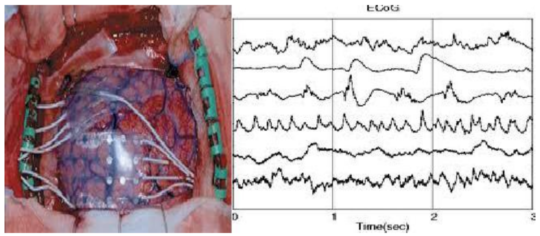
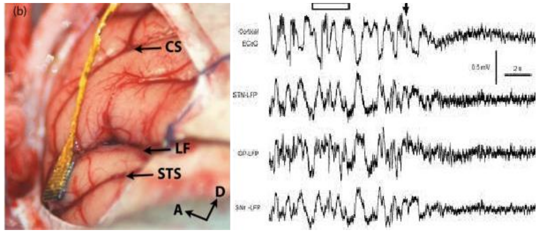
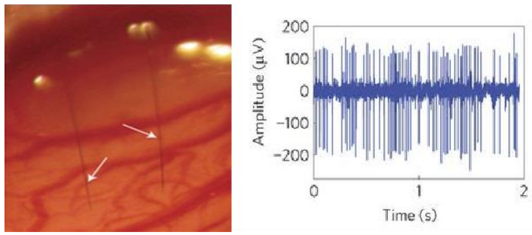


Image Source: http://bio1152.nicerweb.com/Locked/media/ch48/neuron.html

Section 2 in Sornmo and Laguna, Bioelectrical Signal Processing in Cardiac and Neurological Applications, 2005

4.1 Common Brain Signals:

Least Tissue Invasive

Signal		Frequency	Location
EEG		0.1-200Hz	scalp
ECoG		>1kHz	Surface of cortex
LFP		0.1 to 200Hz	Targets neural assemble
Neural Spikes		300Hz-7kHz	Microscopic scale, closer to neuron body

Highest Spatial Resolution

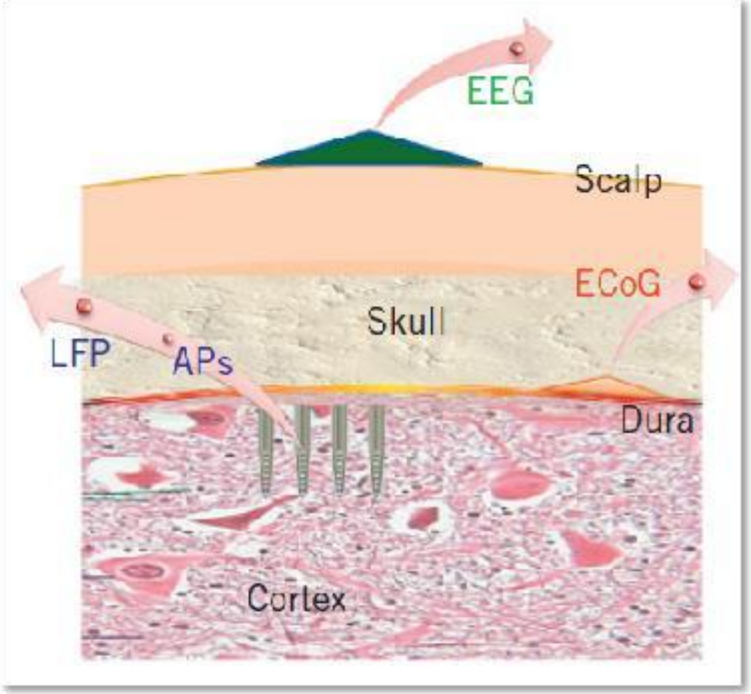


Fig. 2. Location of different Brain Signals

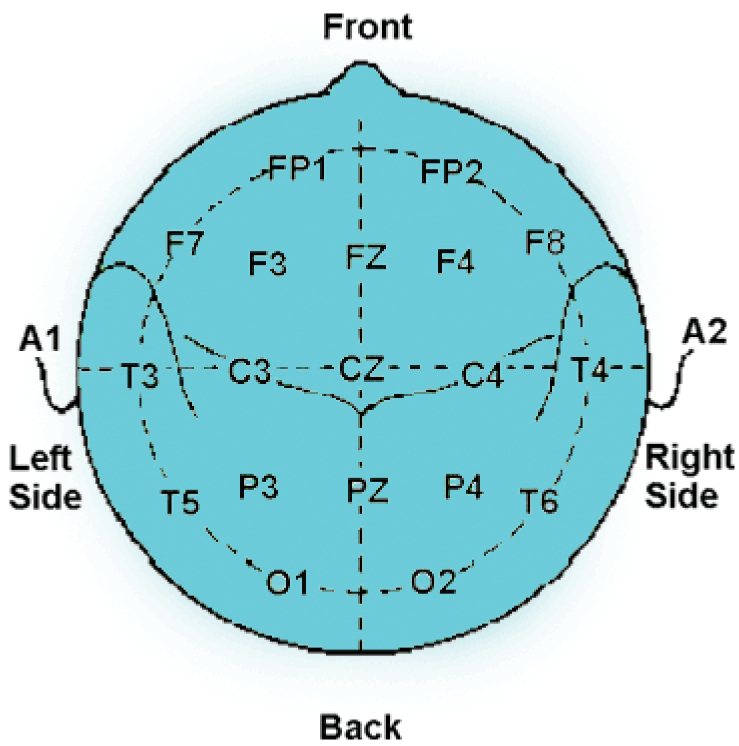
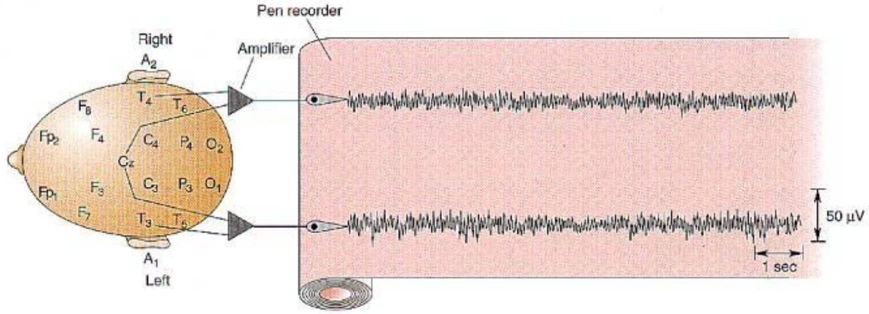


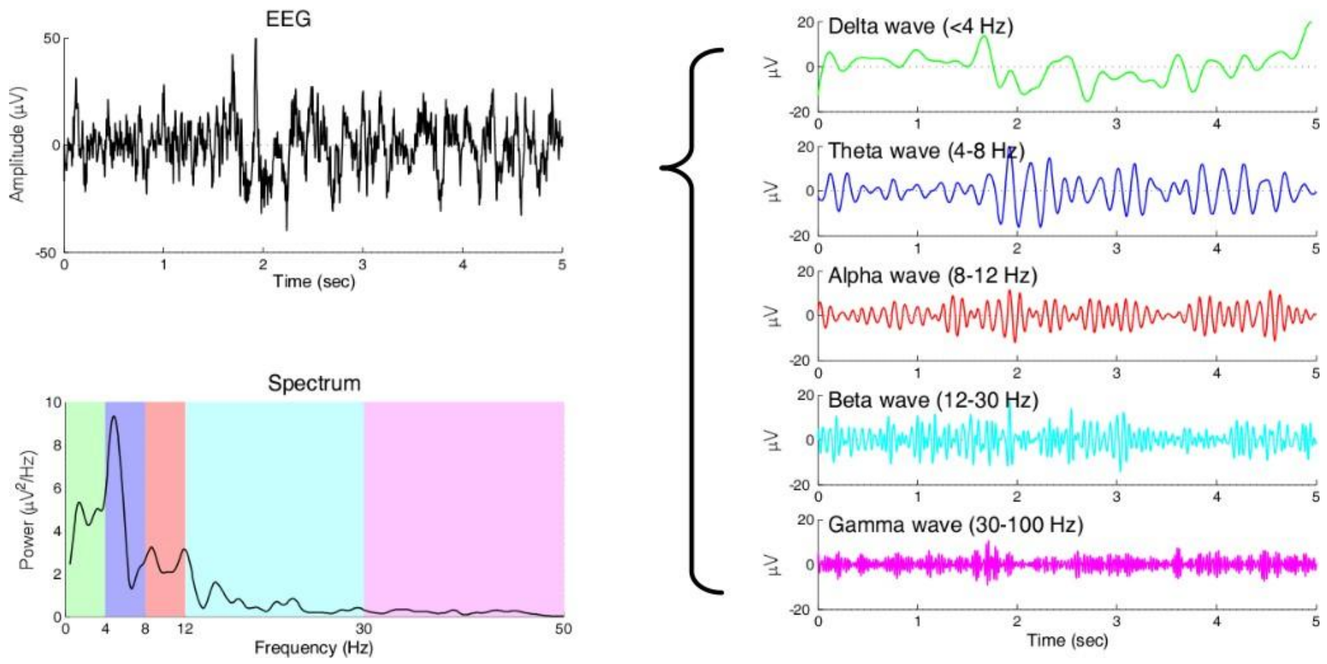
Fig. 3. International 10-20 system

- The International 10-20 system means the actual distances between adjacent electrodes are either 10% or 20% of the total front–back or right–left distance of the skull.
- Original: 19 electrodes. Extension: 32, 64, 128, 256 electrodes, provides higher spatial resolution.

- **EEG signals always represent the potential difference between an active and reference electrodes.**



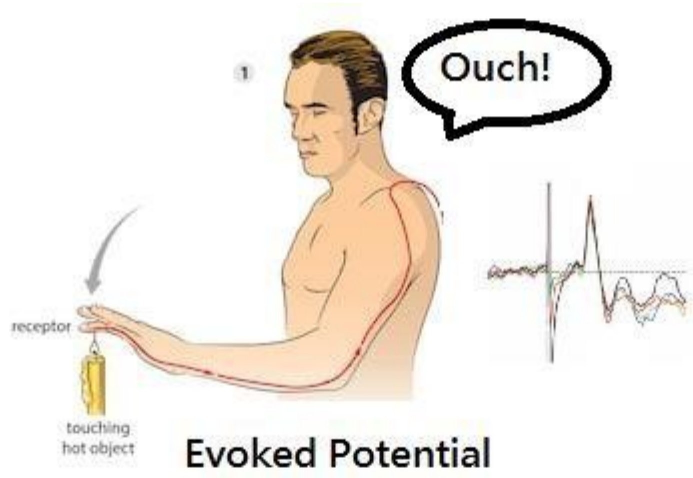
- **Brain waves are made up of different frequencies.**



4.2 General properties of EEG:

- EP (Evoked potential) is a type of ERP (event related potential) generated following the presentation to a stimulus:

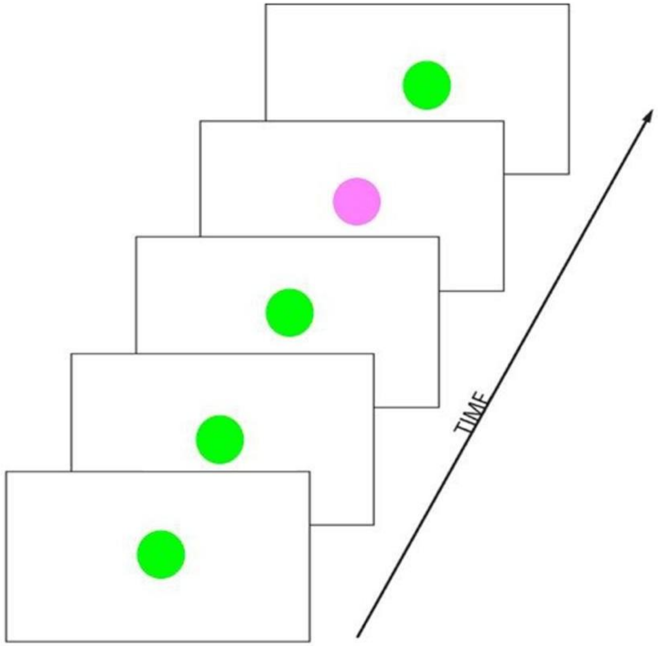
1. Visual, 2. Auditory, 3. olfactory (smell), 4. Somatosensory, 5. Gustatory (taste)



- ERPs belongs to the general family of signals, caused by specific sensory, cognitive, or motor event, some may involve "higher-level" processes, such as memory, attention, expectation, or changes in the mental state.

4.3 Applications:

- **Oddball paradigm:** an experiment in which a subject detects an occasional "target" stimulus in a regular train of standard stimuli
- **P300:** A positive ERP component elicited in an odd ball paradigm



Applications:

Diagnosis, Monitoring, Brain-computer interfa ce

Mind-controlled Keyboard

Mind-controlled Wheelchair

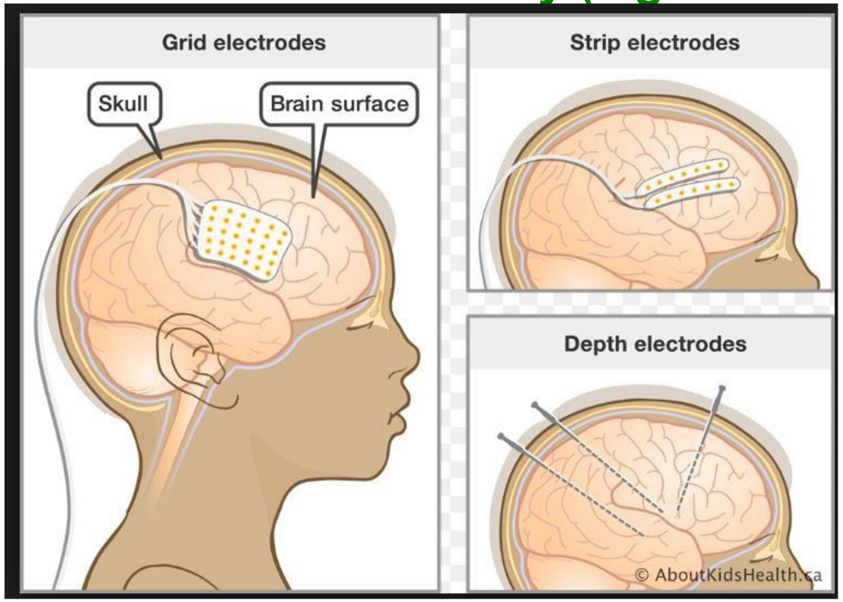
Mind-controlled Robotic Arm

Mind-controlled Walking Robot

5. DIFFICULTIES IN BIOMEDICAL SIGNAL PROCESSING

- **Accessibility:**
  - **Patient safety: prefer non-invasive signal acquisition**
  - **Indirect measurements**
  - **Energy limitation (run out of battery for portable devices)**

- **Interactions among physiological system: cannot isolate multiple factors inside the human body (e.g. brain may perform different functions)**



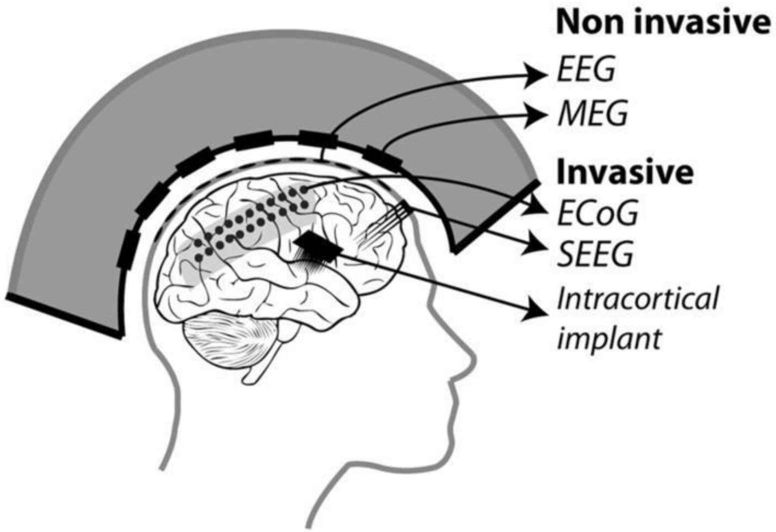
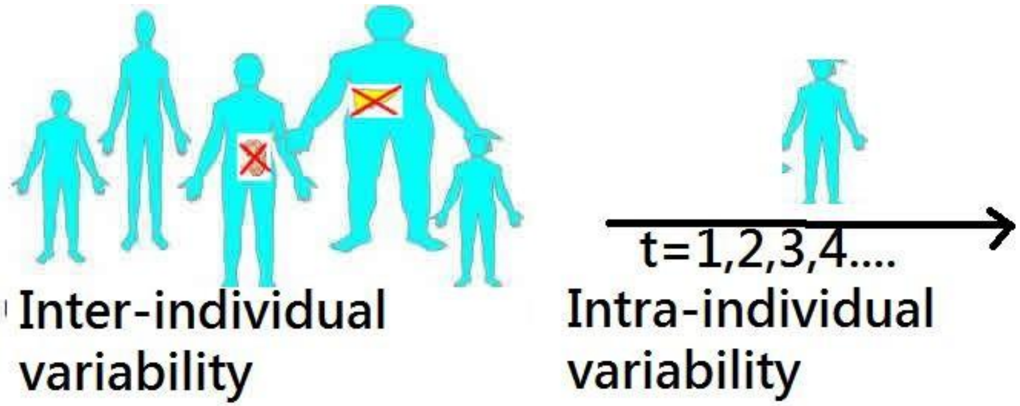


Image source: https://assets.aboutkidshealth.ca/akhassets/intracranial_electrods_EN.png?RenditionID=19

- **Variability**
  - Inter-individual variability: difference occurring in different individual at a specified treatment
  - Intra-individual variability: difference occurring within the same individual when they receive the same treatment at different time/situations

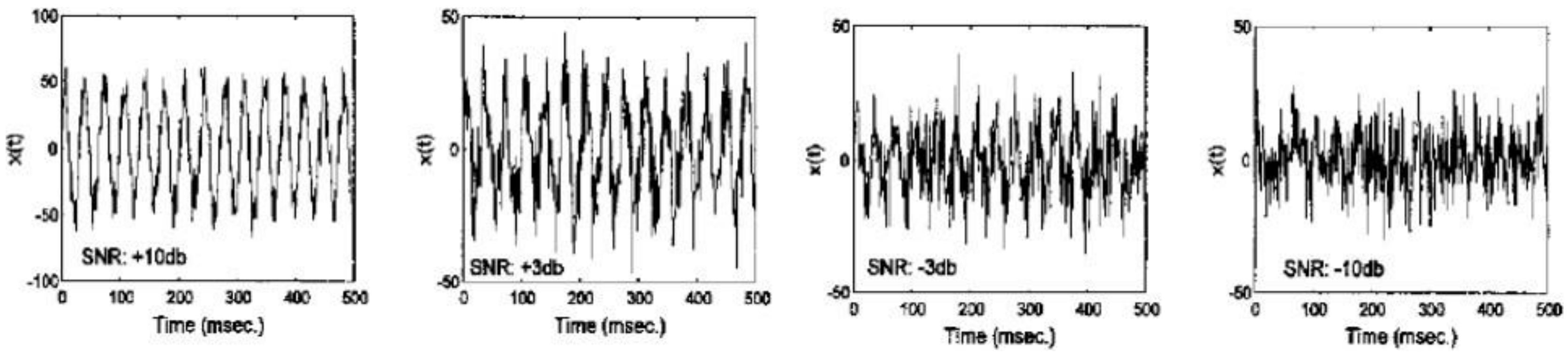
- **Lack of Golden Standard/Ground Truth**
  - There is no ground truth on the internal mechanism of human being



- **Artifact/Interference:**
  - Both are unwanted signals
  - **Signal:** quantity that carries useful information.
  - **Noise:** unwanted quantity which is usually random.

Example:

EMG (Electric potential generated by muscle) and EOG (Electric potential generated from the eye) are meaningful signals but they are regarded as noise or artifacts in EEG processing

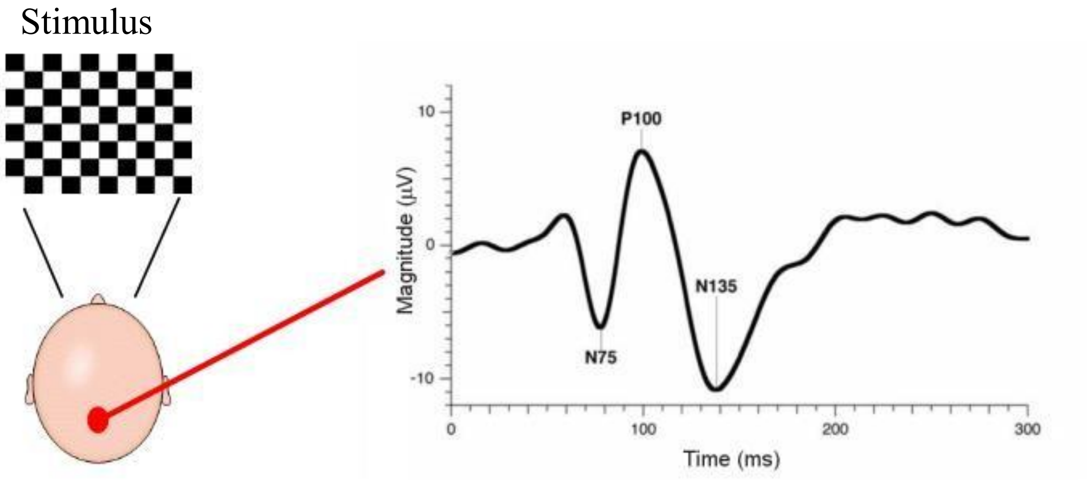


A 30 Hz sine wave with varying amounts of added noise. The sine wave is barely discernable when the SNR is −3dB and not visible when the SNR is −10 dB.

5.1 Noise in Biomedical Signals

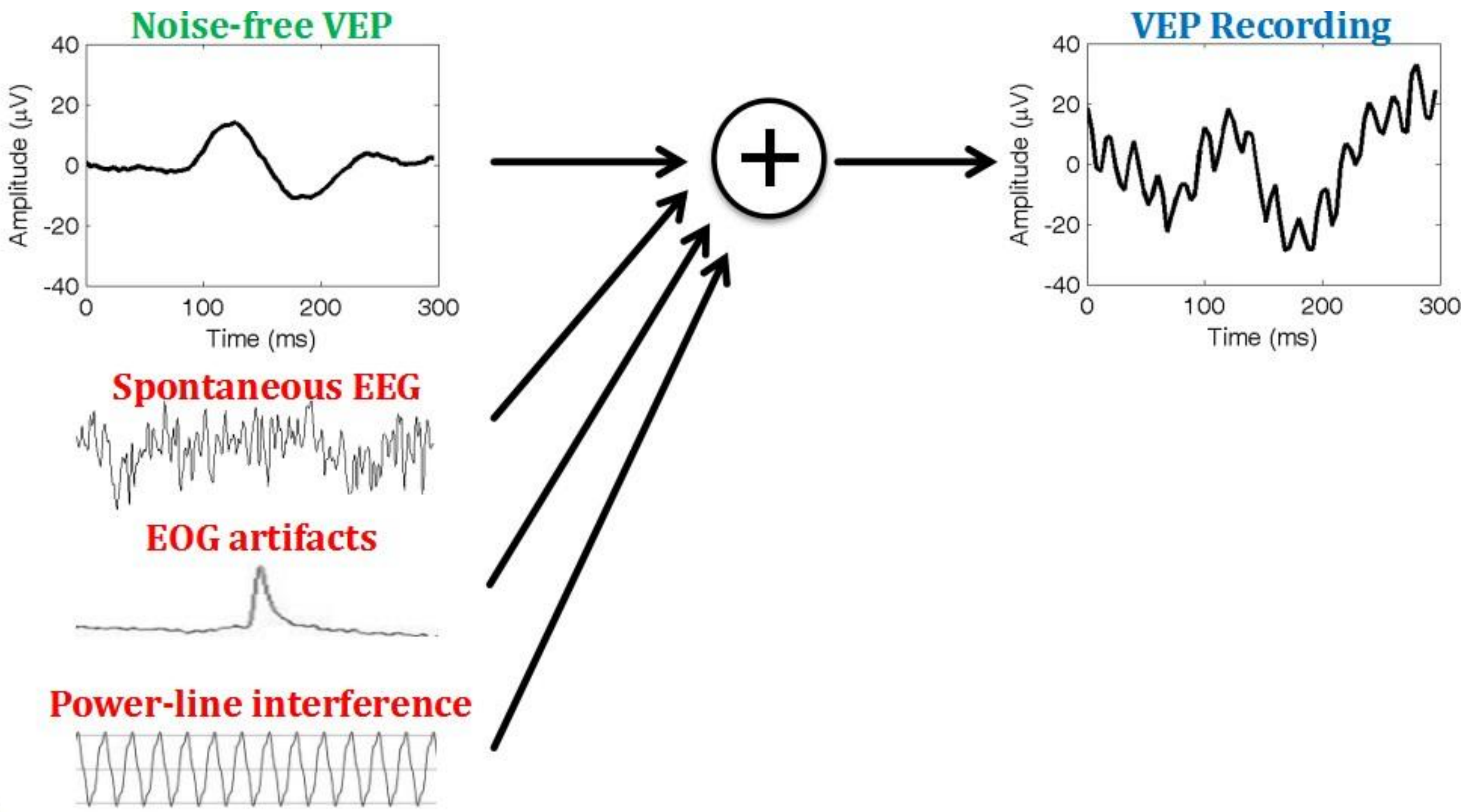
Example: Visual evoked potentials (VEPs)

- **What is EP?: When human eye is presented a stimulus, an electrical potential will be generated from the nervous system, which differs from the normal condition in the EEG.**



- **Visual evoked potential: Electrical potential evoked due to visual stimulation**

VEPs: sources of noise and interferences



5.2 Noise Removal

Methodologies:

1.Filtering in specific domain (on which the signal and the noise are separable)

time domain; frequency domain; spatial domain

2. Pattern recognition and data mining

Challenges:

a) Weak signals:

SNR of ECG: normally >5dB, but SNR of EPs: usually <0dB, because the brain could be occupied by background activities and other unwanted signals, I.e. eye blinks, lid movement

Gamma-band activity in EEG: <0.1% of the total energy

b) Unknown nature of noise: Gaussian assumption may not be true

c) Limited knowledge about information

Ch-1 Introduction (Tutorial Problems)

A. Multiple Choice Questions

1. Which category does OAE belongs to?

(A) Electrical Signals

(B) Magnetic Signals

(C) Acoustic Signals

(D) Optical Signal

2. Which category does MRI belongs to?

(A) Electrical Signals

(B) Magnetic Signals

(C) Acoustic Signals

(D) Optical Signal

3. Which category does PPG belongs to?

(A) Electrical Signals

(B) Magnetic Signals

(C) Acoustic Signals

(D) Optical Signal

4. Which category does ECG belongs to?

(A) Electrical Signals

(B) Magnetic Signals

(C) Acoustic Signals

(D) Optical Signal

5. What does the ECG measure?

(A) Heart Signal

(B) Muscle Signal

(C) Brain Signal

(D) Eye Signal

(E) Ear Signal

6. What does the EOG measure?

(A) Heart Signal

(B) Muscle Signal

(C) Brain Signal

(D) Eye Signal

(E) Ear Signal

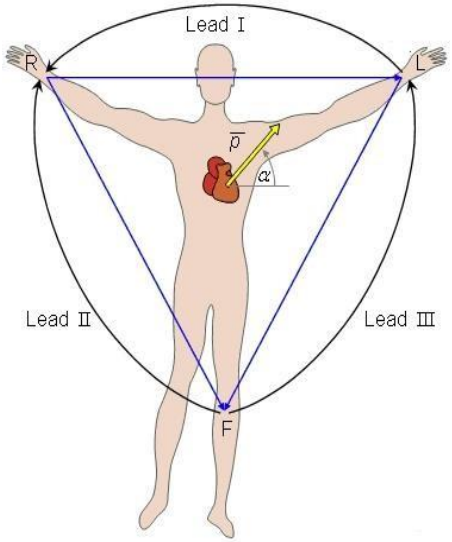
7. Which of the following is NOT a property of the action potential?

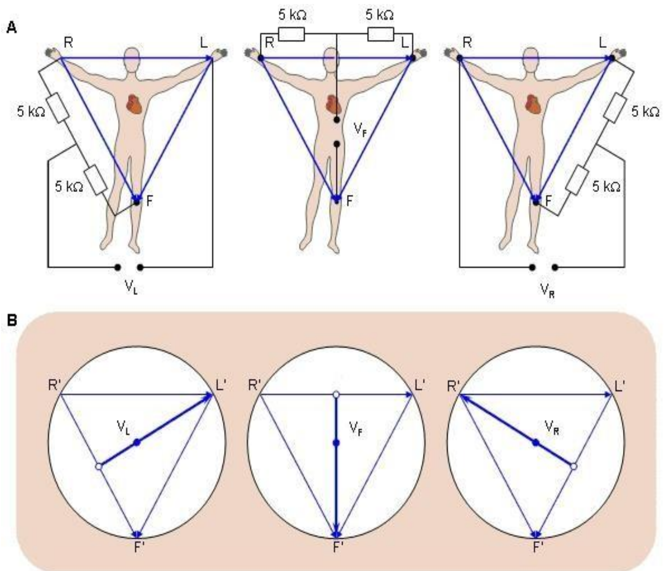
   A. origin of all electrophysiological signals
   B. Measures the Cardiac pace
   C. a transient event in which the electrical membrane potential of a cell rapidly rises and falls, following a consistent trajectory
   D. Caused by flow of ions across cell membrane

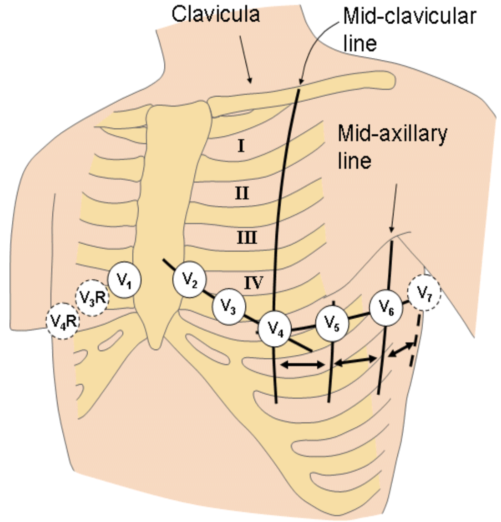
8. Which of the following does the ST segment capture?
   - (A) the sequential depolarization of the right and left atria.
   - (B) the length of a ventricular cardiac cycle
   - (C) the interval during which the ventricles remain in an depolarized state
   - (D) depolarization of the right and left ventricles
9. Which of the following does the QRS complex capture?
   - (A) the sequential depolarization of the right and left atria.
   - (B) the length of a ventricular cardiac cycle
   - (C) the interval during which the ventricles remain in an depolarized state
   - (D) depolarization of the right and left ventricles
10. Which of the following does the P wave capture?
   - (A) the sequential depolarization of the right and left atria.
   - (B) the length of a ventricular cardiac cycle
   - (C) the interval during which the ventricles remain in an depolarized state

(D) depolarization of the right and left ventricles

11. Which of the following diagram illustrates the bipolar limb leads?







(A) (B) (C)

12. Which of the following brain signal is most tissue invasive ?

(A) EEG

(B) ECoG

(C) Neural Spikes

(D) LFP

13. Which of the following brain signal attains the highest spatial resolution?

(A) EEG

(B) ECoG

(C) Neural Spikes

(D) LFP

14. Which of the following is an advantage of using sEMG?

(A) High Spatial Resolution (B) Non-invasive (C) Cannot detect small muscles

15. Which of the following(s) are true?

(1) MRI measures brain structure

(2) fMRI reveals brain function

(3) fMRI measures changes in blood flow

(4) MRI measures brain wave

(A) 1 only (B) 2 only (C) 1,2, and 3 (D) 1,2,3 and 4

16. Which of the following about intra-individual variability is true?

(A) difference occurring in different individual at a specified treatment
(B)difference occurring within the same individual when they receive the same treatment at different time/situations

17. Which of the following(s) are possible sources of noise in VEP?

(1) Spontaneous EEG

(2) EOG artifacts

(3) MUAPs arising from muscles

(4) Power line interference

(A) 1 only (B) 2 only (C) 1,2, and 3 (D) 1,2, and 4

18. Which of the following is NOT a noise removal techniques for biomedical signals?

(A) Filtering in specific domain

(B) Installing magnetic shield

(C) Pattern recognition and data mining

Ans:

1 C	2 B	3 D	4 A	5 A
6 D	7 B	8 C	9 D	10 A
11 A	12 C	13 C	14 B	15 C
16 B	17 D	18 B		